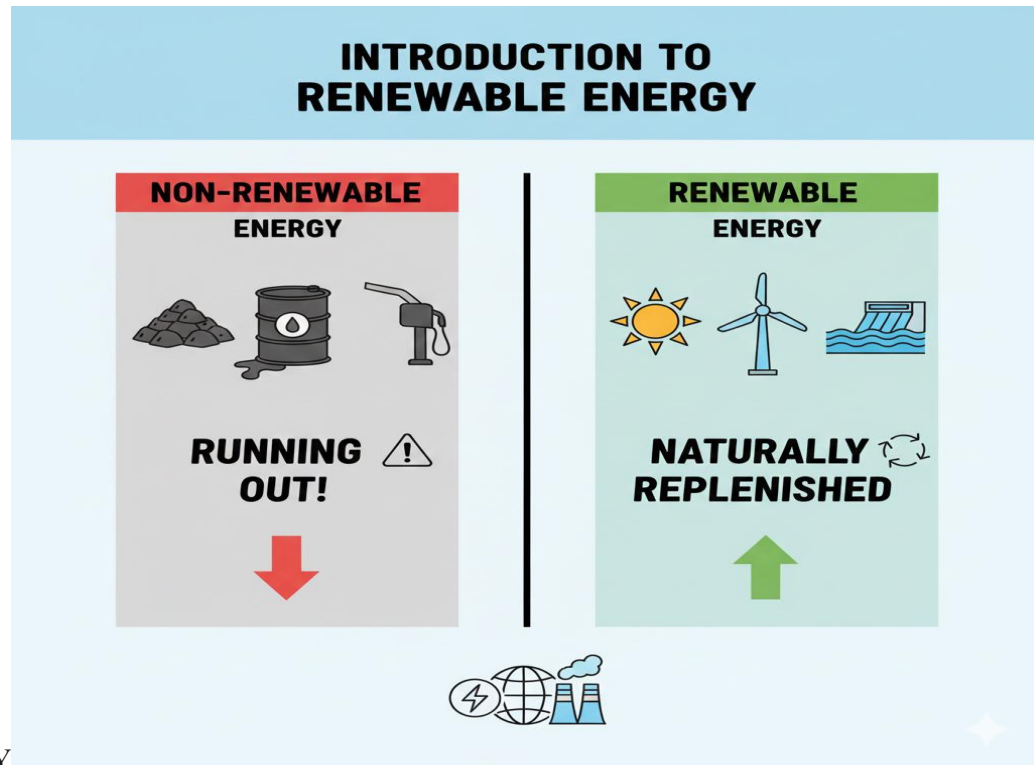


SS1 SOLAR PHOTOVOLTAIC INSTALLATION & MAINTENANCE LESSON NOTES

FIRST TERM

WEEK 1: INTRODUCTION TO RENEWABLE



ENERGY

MEANING OF RENEWABLE AND NON-RENEWABLE ENERGY

Energy is the power needed to do work, provide heat, or generate electricity. The sources of this energy are divided into two main categories:

1. NON-RENEWABLE ENERGY:

- **Definition:** These are energy sources that are finite and will eventually run out. They are formed from the remains of dead plants and animals (fossils) over millions of years.
- **Examples:** Fossil Fuels (Crude Oil, Coal, Natural Gas), Nuclear Fission (Uranium).
- **Impact:** When burned, they release harmful greenhouse gases (like carbon dioxide, CO_2), which contribute to air pollution and global climate change.

2. RENEWABLE ENERGY (RE):

- **Definition:** These are energy sources that are **naturally replenished** or "renewed" on a human timescale.⁶ They are considered "clean" because they produce little to no greenhouse gas emissions.⁷
- **Examples:** Solar, Wind, Hydro, Biomass, Geothermal.

SOURCES OF RENEWABLE ENERGY

- **SOLAR:** Energy derived directly from the **sun's radiation**. This is the most abundant energy source on Earth. It can be captured in two ways:
 1. **Photovoltaic (PV):** Using solar panels to convert sunlight directly into electricity.
 2. **Solar Thermal:** Using the sun's heat to heat water or create steam to turn a turbine.
- **WIND:** Energy derived from the **movement of air (wind)**. Large wind turbines (propellers) are turned by the wind, which spins a generator to create electricity.
- **HYDRO:** Energy derived from the **movement of water**. This is typically done by building a **dam** on a large river. The water held behind the dam is released, flowing through a turbine to generate electricity (e.G., the Kainji Dam in Nigeria).
- **BIOMASS:** Energy derived from **organic matter** (plants or animal waste).
- **Examples:** Burning wood for cooking, fermenting crops (like corn or sugarcane) to create biofuels (ethanol), or processing animal manure in a "biodigester" to create methane gas for cooking.

IMPORTANCE OF SOLAR ENERGY IN NIGERIA

Solar energy is arguably the most important renewable energy source for Nigeria's future, for several critical reasons:

1. **ABUNDANT NATURAL RESOURCE:**
 - Nigeria is located in the "high-sunshine belt" (tropics) and receives an average of **5.5 to 7.0 Peak Sun Hours (PSH)** per day. This is a massive, untapped natural resource that is more reliable than wind or hydro in many parts of the country.
2. **GRID INSTABILITY AND ENERGY DEFICIT:**

- The national grid is often unreliable, with frequent power cuts ("NEPA failure") and load shedding. This forces businesses and homes to rely on expensive, noisy, and polluting **diesel or petrol generators**. Solar power provides a clean, quiet, and increasingly cheaper alternative.

3. **RURAL ELECTRIFICATION:**

- A large portion of Nigeria's rural population is not connected to the national grid.¹⁸ Extending the grid to these remote areas is extremely expensive.
- **Solar off-grid systems** (see Week 4) are the perfect solution. They can be installed "a-la-carte" for a single home, a school, or a village, providing power for lights, phone charging, and small businesses without needing the grid.

4. **LOW OPERATING COSTS:**

- While the initial cost of a solar system can be high, its "fuel" (sunlight) is **100% free**. Once installed, the maintenance is minimal, and the operating cost is almost zero, unlike generators that require constant refueling.

5. **ENVIRONMENTAL BENEFITS:**

- Relying on solar power reduces the carbon emissions from generators, improving air quality and public health.

6. **ECONOMIC GROWTH:**

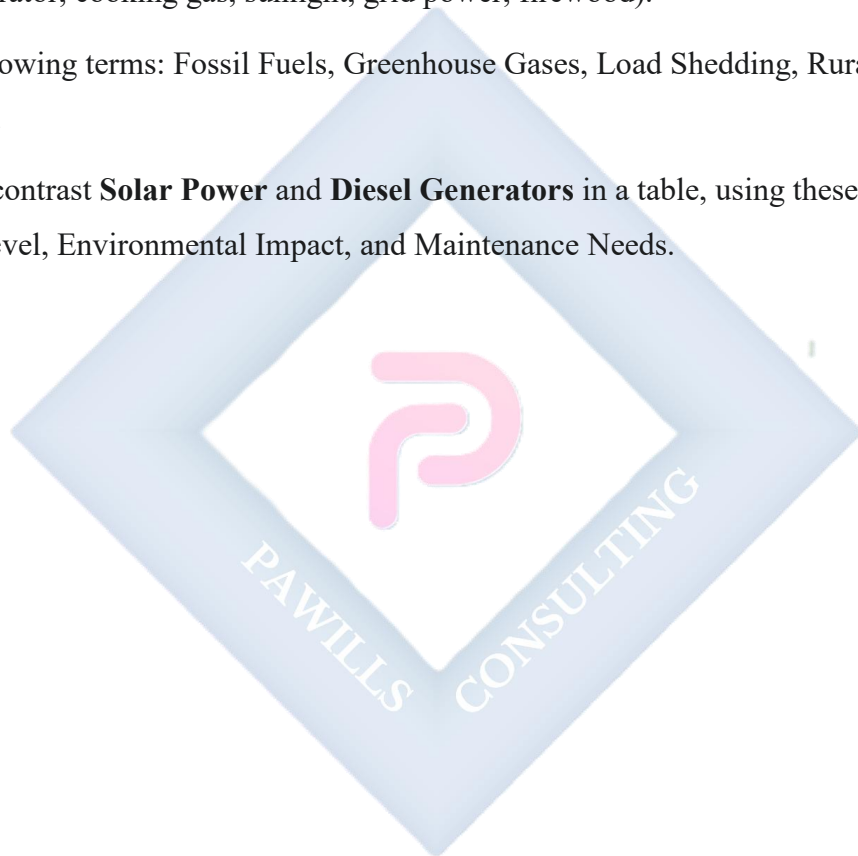
- A stable power supply allows businesses to operate, clinics to function, and students to study. The growth of the solar installation industry itself creates thousands of "green jobs" for technicians, installers, and engineers.

EVALUATION

1. What is the main difference between a renewable energy source and a non-renewable energy source?
2. List four examples of renewable energy and two examples of non-renewable energy.
3. Explain the two primary ways solar energy can be captured (PV and Thermal).
4. What is "biomass" energy? Give two local examples.
5. List and explain three specific reasons why solar energy is critically important for Nigeria's development.

ASSIGNMENT

1. Write a 200-word essay on the topic: "The Impact of Unreliable Grid Power on Daily Life and Business in Nigeria."
2. Research the Kainji Dam. What type of renewable energy does it use, and what is its generating capacity?
3. Observe your local community. List all the different sources of energy you see being used in one day (e.g., generator, cooking gas, sunlight, grid power, firewood).
4. Define the following terms: Fossil Fuels, Greenhouse Gases, Load Shedding, Rural Electrification.
5. Compare and contrast **Solar Power** and **Diesel Generators** in a table, using these criteria: Fuel Cost, Noise Level, Environmental Impact, and Maintenance Needs.

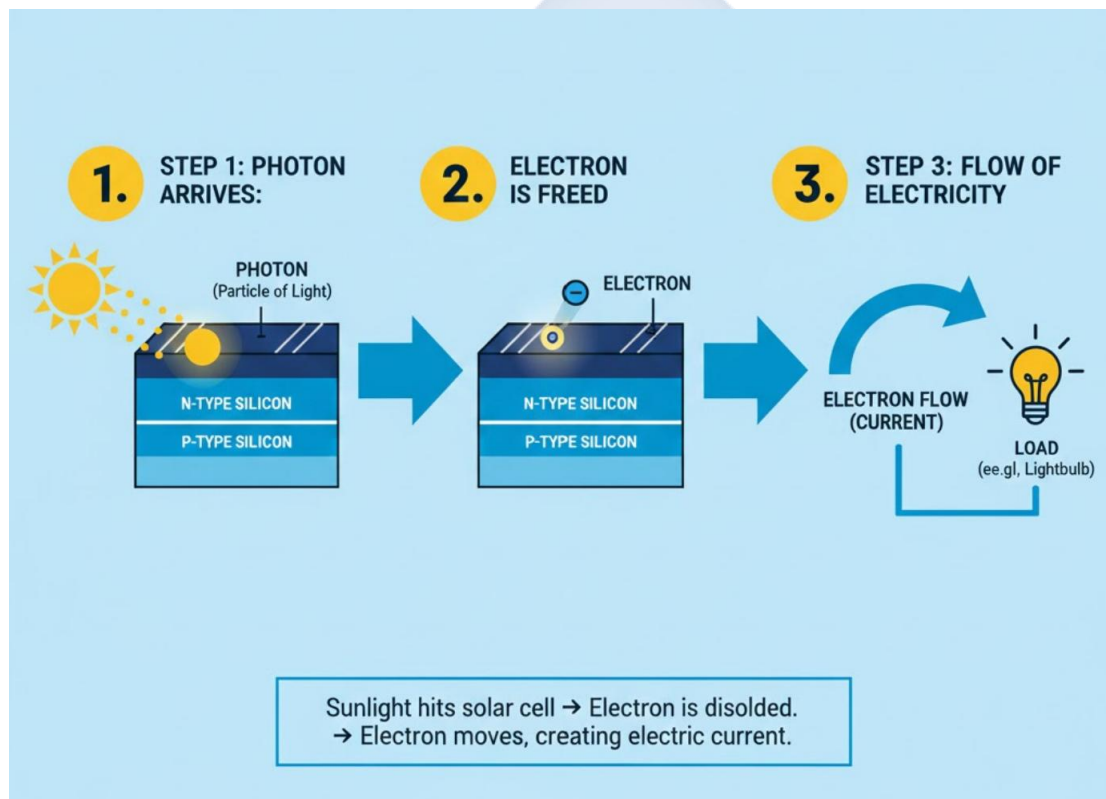


WEEK 2: BASICS OF SOLAR ENERGY

SOLAR RADIATION

Solar Radiation is the energy (or light) that the sun emits. This energy travels 150 million kilometers to Earth in the form of **photons**. Photons are tiny packets of energy.

- This radiation covers a wide electromagnetic spectrum, including visible light, ultraviolet (UV) light, and infrared (IR) light.
- The **photovoltaic effect** primarily uses the **visible light** portion of the spectrum to generate electricity.



THE PHOTOVOLTAIC (PV) EFFECT

This is the most important concept in this course. The **Photovoltaic (PV) Effect** is the scientific process that converts solar radiation (sunlight) directly into electricity.

This process occurs inside a **solar cell**, which is made of a **semiconductor** material, usually **silicon**.

Here is a step-by-step breakdown:

1. **STRUCTURE:** A solar cell is made of two layers of silicon (a P-type layer and an N-type layer) sandwiched together, creating an electric field called a "P-N junction."
2. **IMPACT:** A **photon** (from sunlight) with enough energy strikes the surface of the solar cell.
3. **RELEASE:** This photon's energy is absorbed, and it **knocks an electron "loose"** from its atom in the silicon.
4. **MOVEMENT:** This free electron is pushed by the electric field (the P-N junction) to move to one side of the cell. This leaves a "hole" (the absence of an electron) which moves to the other side.
5. **FLOW (ELECTRICITY):** This separation of electrons (negative charge) and holes (positive charge) creates a **voltage** (electrical pressure).
6. **THE CIRCUIT:** When we connect these two sides (front and back of the cell) with a wire, the loose electrons flow through the wire to get back to the holes. This flow of electrons is what we call **electricity** or an **electric current**.
 - The electricity produced by a solar panel is **Direct Current (DC)**.

ADVANTAGES AND LIMITATIONS OF SOLAR ENERGY

ADVANTAGES

1. **RENEWABLE & ABUNDANT:** The sun is a massive, inexhaustible source of energy. It will not run out for billions of years.
2. **CLEAN & ENVIRONMENTALLY FRIENDLY:** Solar panels produce **zero emissions** (no ³³\$CO_2\$, no fumes) while operating. They are silent and do not pollute the air or water.
3. **LOW OPERATING & MAINTENANCE COSTS:** Once installed, the fuel (sunshine) is free. Maintenance is minimal—mostly just cleaning the panels.
4. **SCALABLE & VERSATILE:** You can have a tiny system to charge a phone (a solar lantern) or a massive system to power a factory (a "solar farm").
5. **ENERGY INDEPENDENCE:** A home or business with a solar system (especially with batteries) is no longer helpless when the national grid fails.

6. **GRID REDUCTION:** Even for homes with grid power, a solar system can drastically reduce monthly electricity bills.

LIMITATIONS

1. **INTERMITTENT:** Solar panels only produce power when the sun is shining. They do not work at night and have reduced output on very cloudy or rainy days.
2. **HIGH INITIAL COST:** The "upfront" cost of buying the panels, batteries, and inverter can be very high, even though it saves money in the long run.
3. **STORAGE REQUIREMENT:** Because it's intermittent, you *must* have **batteries** (which are expensive and have a limited lifespan) if you want power at night. This is the main challenge of **off-grid** systems.
4. **SPACE INTENSIVE:** You need a large, shadow-free area (usually a roof) to install enough panels to produce a significant amount of power.
5. **EFFICIENCY:** Solar panels are not 100% efficient. Most commercial panels can only convert 17-22% of the sunlight they receive into electricity.

EVALUATION

1. Define the term "Solar Radiation."
2. What is a "photon"?
3. In your own words, explain the Photovoltaic (PV) Effect step-by-step.
4. What type of electricity (AC or DC) does a solar panel produce?
5. List three major advantages and three major limitations of using solar energy.

ASSIGNMENT

1. Write a 250-word essay explaining the Photovoltaic Effect. You must use the following key terms: **Photon, Silicon, Electron, P-N Junction, DC Current.**
2. Draw a simple, labeled diagram showing a photon striking a solar cell, knocking an electron loose, and creating an electrical current.

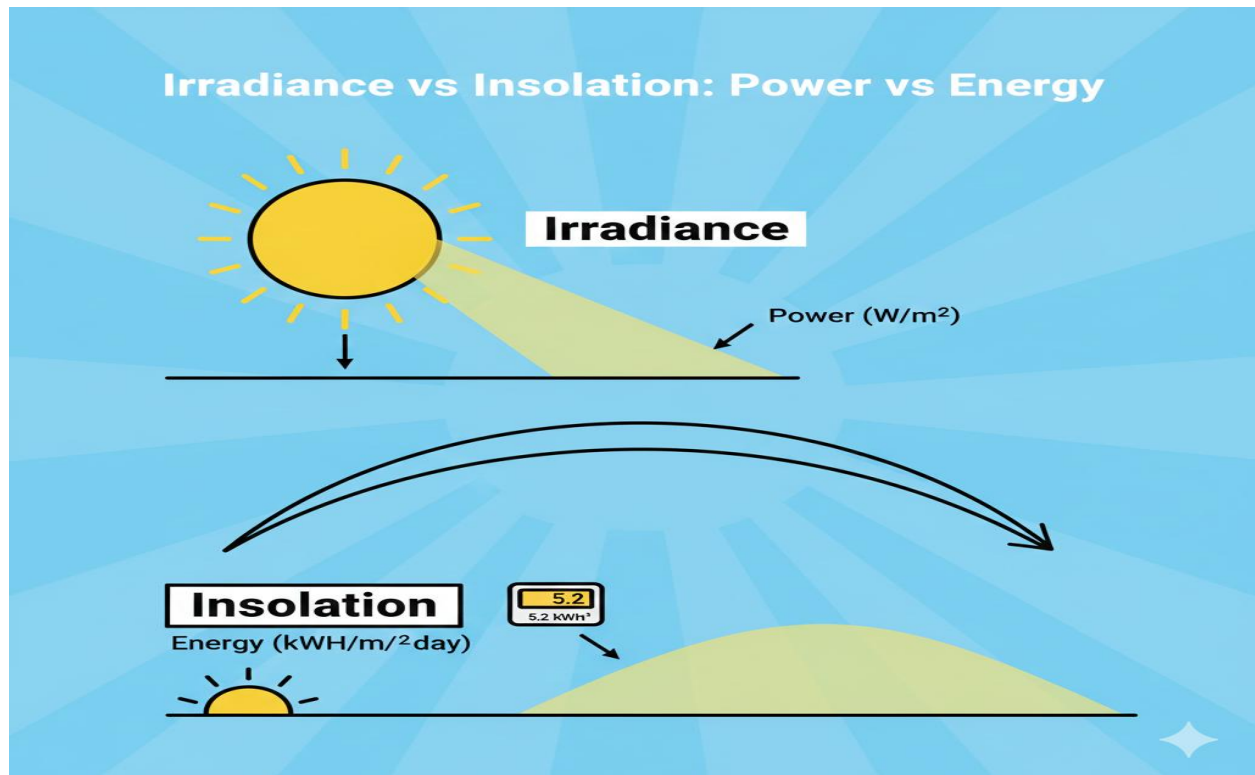
3. Research the main material used in solar cells: **Silicon**. Why is it used? (Hint: It is a semiconductor).
4. Which "limitation" of solar energy do you think is the *biggest* problem for a typical homeowner in Nigeria? The initial cost, or the intermittency (needing batteries)? Justify your answer.
5. Define the following terms: Semiconductor, Direct Current (DC), Intermittent, Scalable.



WEEK 3: SOLAR ENERGY TERMINOLOGIES

INTRODUCTION

To install, size, and maintain a solar system, you must speak the language. This week, we cover the essential technical terms and units used to measure solar energy. Understanding these is the difference between a system that works and one that fails.



IRRADIANCE (INSTANTANEOUS POWER)

- **DEFINITION:** Solar **Irradiance** is the amount of **power** from the sun that lands on a specific area (like one square meter) at a *single instant in time*.
- **ANALOGY:** Think of it as the "**brightness**" or "**intensity**" of the sun *right now*. At noon, the irradiance is high. Near sunset, it is low.
- **UNIT:** Watts per square meter (W/m^2).
- **STANDARD VALUE:** A solar panel's power (e.g., 300W) is rated using a "Standard Test Condition" (STC) irradiance of **1000 W/m^2** . This is the "peak" brightness.

INSOLATION (ENERGY OVER TIME)

- **DEFINITION:** Solar **Insolation** is the total amount of solar **energy** (not power) that lands on a specific area over a *period of time* (usually one day).
- **ANALOGY:** If irradiance is the *speed* you are driving, insolation is the *total distance* you drove all day. It's the *sum* of all the irradiance you received.
- **UNIT:** Kilowatt-hours per square meter per day (kWh/m²/day).

PEAK SUN HOURS (PSH)

- **DEFINITION:** This is the most important term for system sizing. **Peak Sun Hours (PSH)** is a standard value that represents the *equivalent* number of hours per day that the sun was shining at its *peak irradiance* (1000 W/m²).
- **EXPLANATION:**
 - In the real world, the sun is weak in the morning, strong at noon, and weak in the evening.
 - PSH "averages" this out. It takes all the weak and strong sunlight from the whole day and repackages it as a *single number*.
 - If a location gets **5 PSH**, it means the *total energy* it received all day is *equal* to what it would have received if the sun was at its peak (1000 W/m²) for **5 straight hours**.
- **APPLICATION (SIZING):** This number tells you how much energy your panel will *actually* produce in a day.
 - **Formula:** Daily Energy (Wh) = Panel's Power (W) × Peak Sun Hours (PSH)
 - Example: A 200W panel in a location with 5 PSH will produce:

200 Watts × 5 Hours = 1000 Watt-hours (Wh) or 1 kWh of energy per day.

- **NIGERIA'S VALUE:** As we learned, Nigeria has excellent PSH, generally between **5.5 and 7.0 PSH**, depending on the region (higher in the North).

EFFICIENCY

- **DEFINITION:** Solar panel **efficiency** is the ratio (as a percentage) of the electrical power a panel *produces* compared to the solar power it *receives*.
- **ANALOGY:** It's the panel's "pass mark." If 1000W of sunlight (irradiance) hits a 1 m² panel, and the panel only produces 200W of electricity, its efficiency is 20%.
- **FORMULA:** Efficiency (%) = (Electrical Power Out / Solar Power In) × 100
- **TYPICAL VALUES:**
 - Monocrystalline Panels: 18-23% (High)
 - Polycrystalline Panels: 15-18% (Medium)
 - Thin-Film Panels: 10-13% (Low)
- **WHY IT MATTERS:** Higher efficiency means you get *more power* from a *smaller panel*. If your roof space is small, you *must* use high-efficiency panels.

EVALUATION

1. What is the key difference between Solar *Irradiance* and Solar *Insolation*?
2. What is the standard unit of measurement for (a) Irradiance and (b) Insolation?
3. Define "Peak Sun Hours (PSH)."
4. A 300-Watt solar panel is installed in Jos, which has an average of 6.0 PSH. How much energy (in Watt-hours) will this panel produce in one day?
5. What does it mean if a solar panel has an "efficiency of 20%"?

ASSIGNMENT

1. Write a 200-word essay explaining the relationship between Irradiance, Insolation, and Peak Sun Hours. Use an analogy (like filling a bucket with a hose) to explain it.
2. Research and find the average PSH for three different Nigerian cities (e.g., Lagos, Abuja, Kano).
3. You need to generate 2,000 Wh (2 kWh) of energy per day. Your location has 5 PSH. What *total wattage* of solar panels do you need? (Show your calculation).

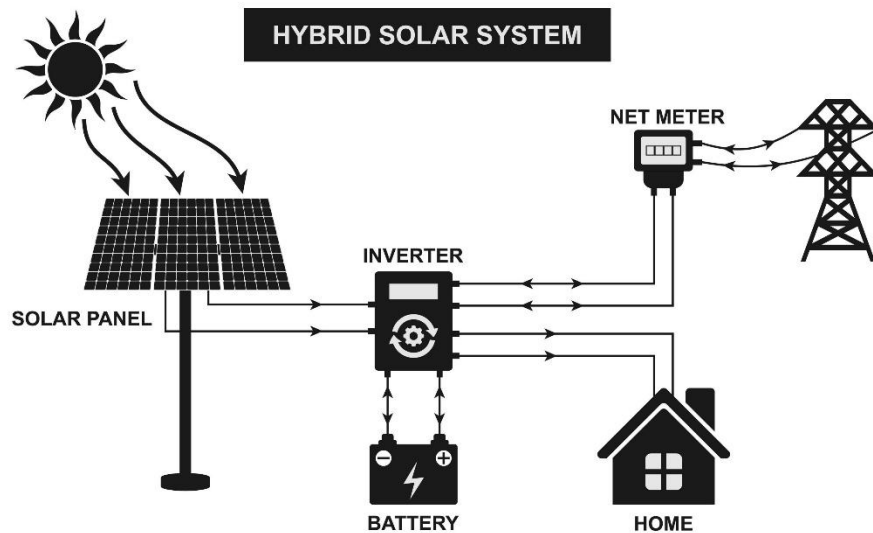
4. A 1.5 m^2 solar panel is tested. It receives 1000 W/m^2 of irradiance (total 1500 W of solar power). It produces 270 W of electrical power. What is the efficiency of this panel? (Show your calculation).
5. Define the following terms: Standard Test Condition (STC), W/m^2 , $\text{kWh/m}^2/\text{day}$, Efficiency.



WEEK 4: TYPES OF SOLAR ENERGY SYSTEMS

INTRODUCTION

A "PV system" is not just the panels. It's an entire system of components working together. The *arrangement* of these components determines the *type* of system. The three types are defined by their relationship to the **utility grid** (e.g., NEPA/PHCN).



OFF-GRID (STAND-ALONE) SYSTEMS

- **DEFINITION:** An **off-grid** system is a PV system that is **completely disconnected** from the utility grid. It is 100% independent.
- **APPLICATION:** This is the *only* option for remote or rural areas with no grid access. It is also used for critical loads that must *never* fail (e.g., remote telecoms towers, military outposts).

KEY COMPONENTS (BLOCK DIAGRAM):

Solar Panels → Charge Controller → Battery Bank → Inverter → AC Loads (Appliances)

- **HOW IT WORKS:**
 1. **Day:** Solar panels produce DC electricity.
 2. This DC electricity goes to the **Charge Controller**.
 3. The Charge Controller *first* uses the power to charge the **Battery Bank**.

4. At the same time, the **Inverter** draws power from the batteries, converts it from DC to AC, and powers the house.
5. **Night:** The panels shut down. The house runs entirely on the energy stored in the **Battery Bank**.
- **REQUIREMENT:** This system **MUST** have a battery bank that is large enough to store energy for night-time use and for cloudy days (called "days of autonomy"). This makes it the most expensive system.

ON-GRID (GRID-TIED) SYSTEMS

- **DEFINITION:** An **on-grid** system is a PV system that is **connected directly to the utility grid**. It does **NOT** have batteries.
- **APPLICATION:** This is the most common system in countries with a *stable* grid (e.g., Germany, USA). Its only purpose is to **reduce the customer's electricity bill**.

KEY COMPONENTS (BLOCK DIAGRAM):

Solar Panels → Grid-Tied Inverter → Utility Grid / AC Loads

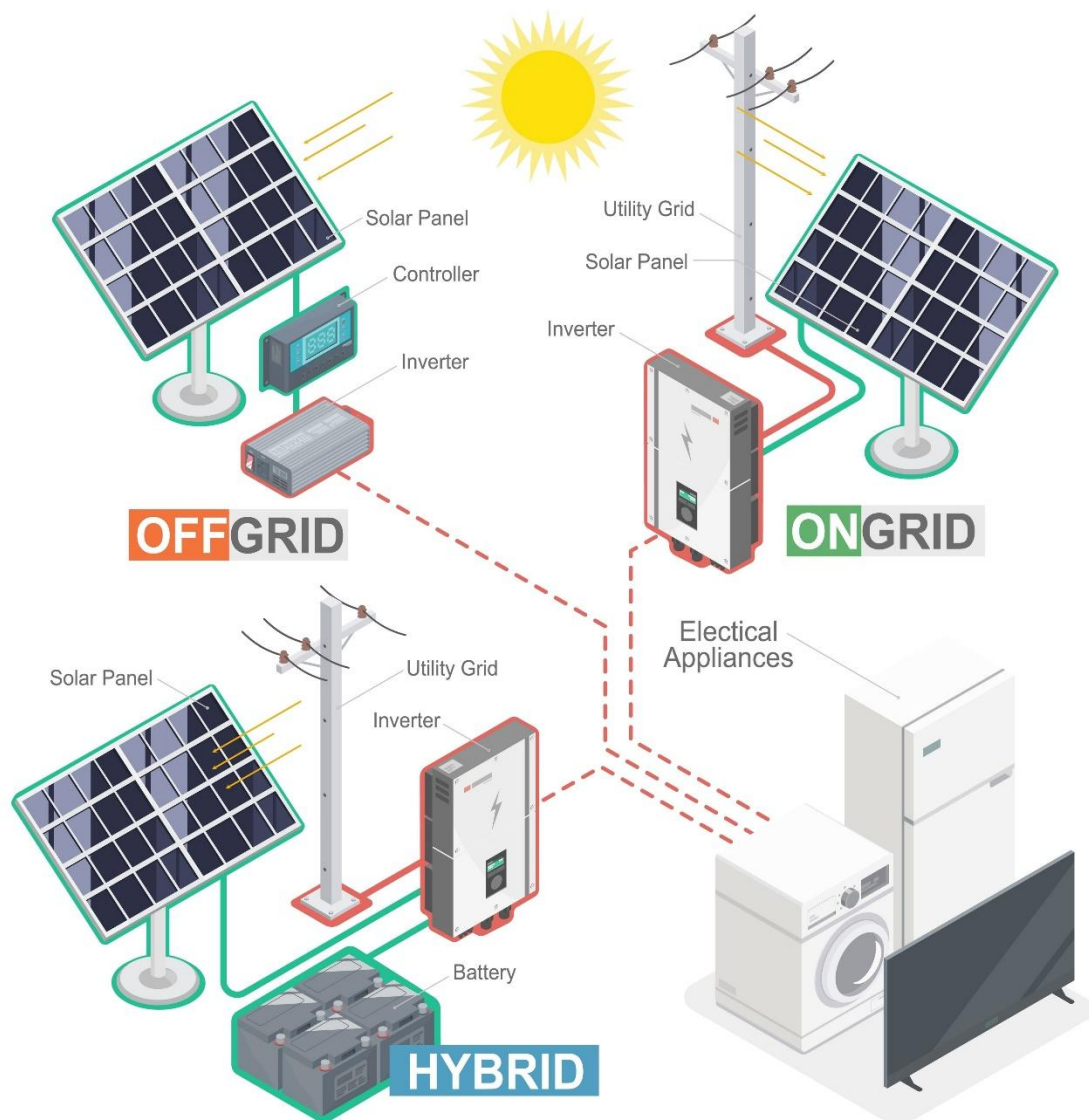
- **HOW IT WORKS:**
 1. **Day:** Solar panels produce DC electricity.
 2. This DC electricity goes to the **Grid-Tied Inverter**.
 3. The inverter converts it to AC *and* synchronizes it with the grid's AC.
 4. The house **first uses the solar power**. If the house needs *more* power, it "pulls" the extra from the grid.
 5. If the panels produce *more* power than the house needs, the excess power is "**sold**" **back to the grid** (this is called **Net Metering**).
 6. **Night:** The panels shut down. The house runs 100% on power from the grid.
- **CRITICAL FLAW:** If the utility grid fails ("NEPA takes light"), the on-grid inverter is *legally required* to shut down (this is called "anti-islanding" and is a safety feature). This means: **If the grid is down, an on-grid system does NOT provide power.**

HYBRID SYSTEMS

- **DEFINITION:** A **hybrid** system is the "best of both worlds." It is **connected to the utility grid** AND it has a **Battery Bank**.
- **APPLICATION:** This is the **ideal (but most expensive) system for the Nigerian context**. It provides bill reduction *and* backup power.
- **KEY COMPONENTS (BLOCK DIAGRAM):**

Solar Panels → Hybrid Inverter → Battery Bank → AC Loads / Utility Grid





• HOW IT WORKS (SMART PRIORITIES):

1. **Day:** Solar panels produce power. The **Hybrid Inverter** (which is a smart inverter, charge controller, and grid-tie all in one) directs the power in a priority order:
 - **Priority 1:** Power the house loads directly.
 - **Priority 2:** Use excess solar to charge the batteries.
 - **Priority 3:** If loads are powered and batteries are full, sell the final excess to the grid (if net metering is available).

2. **Night (Grid ON):** The inverter runs the house from the battery. If the battery gets low, it can recharge from the grid.
3. **Night (Grid OFF):** The inverter automatically disconnects from the grid and runs the house 100% from the battery (like an off-grid system).

EVALUATION

1. What is the main physical difference between an On-Grid system and an Off-Grid system? (Hint: A key component).
2. What is the *only* purpose of an On-Grid (Grid-Tied) system?
3. Why is a standard On-Grid system not a good solution for the "backup power" problem in Nigeria?
4. Explain "Net Metering."
5. Why is a Hybrid system considered the "best of both worlds," especially for a Nigerian home?

ASSIGNMENT

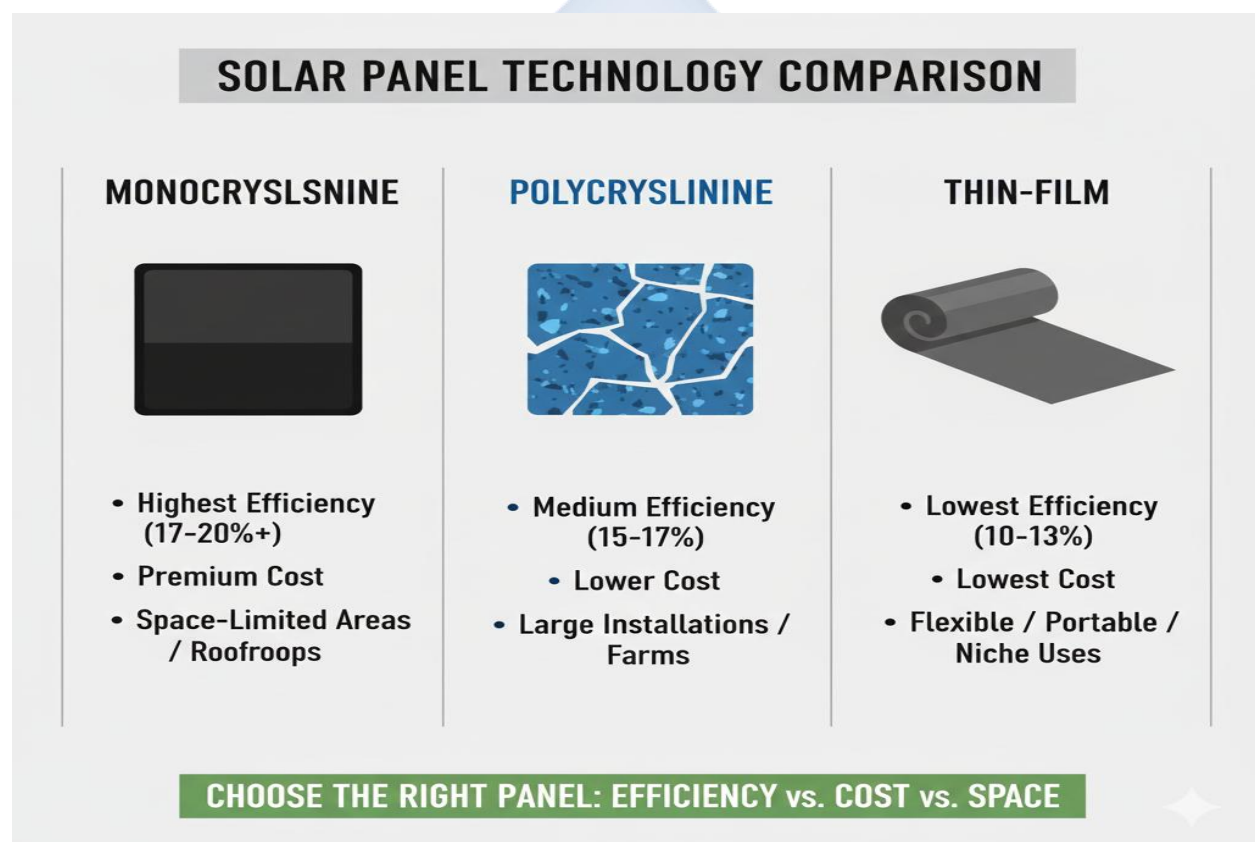
1. Draw a simple, labeled block diagram for all three system types: Off-Grid, On-Grid, and Hybrid.
2. Write a 250-word essay explaining why a simple **Off-Grid** system is the best choice for a rural clinic that has never had "NEPA" light.
3. Write a 250-word essay explaining why a **Hybrid** system is the best choice for a bank in a city like Lagos that *has* grid power but needs 24/7 reliability.
4. Define the following terms: Utility Grid, Net Metering, Days of Autonomy, Anti-Islanding, Hybrid Inverter.
5. Research the company "Tesla." What is their battery product called, and which of the three system types does it enable?

WEEK 6: COMPONENTS OF PV SYSTEMS I (SOLAR PANELS)

INTRODUCTION

The **Solar Panel**, or **Photovoltaic (PV) Module**, is the most visible component of a solar system. It is the "engine" that captures sunlight. A "module" is one single panel. An "array" is a group of modules connected together.

A single **solar cell** produces only ~0.5 Volts. To be useful, cells are connected in **series** (to build voltage) and **parallel** (to build current) inside a weatherproof aluminum frame.



TYPES OF SOLAR PANELS

There are three main types of solar panels, based on the *type of silicon cell* used.

1. MONOCRYSTALLINE PANELS

- **MANUFACTURE:** Made from a *single, pure crystal* of silicon (mono = one). The silicon ingot is grown into a single large crystal and then sliced into "wafers."
- **APPEARANCE:** The cells have a uniform, **dark black** color and **octagonal (cut) corners**.
- **PROS:**
 - **Highest Efficiency:** Typically 18-23%. They produce the most power for a given amount of space.
 - **Best Heat Tolerance:** They perform slightly better in high temperatures.
 - **Longest Lifespan.**
- **CONS:**
 - **Most Expensive:** The manufacturing process is complex and wastes silicon (from cutting the round ingot into "octagons").
- **APPLICATION:** Ideal for **residential roofs** where space is limited and you need maximum power.

2. POLYCRYSTALLINE PANELS

- **MANUFACTURE:** Made from *multiple silicon crystals* (poly = many). Molten silicon is poured into a square mold and cooled, forming many small crystals.
- **APPEARANCE:** The cells have a "marbled" or "flaked" **blue** color. They are perfectly **square**.
- **PROS:**
 - **Cheaper:** The manufacturing process is simpler and wastes less silicon.
 - **Good Performance:** Efficiency is very good, just slightly lower than mono.
- **CONS:**
 - **Lower Efficiency:** Typically 15-18%. You need *more* roof space to get the *same* amount of power as a mono system.
 - Slightly less tolerant of high heat.
- **APPLICATION:** The "workhorse" of the industry. Good for large **commercial roofs** or **ground-mount solar farms** where space is not an issue and cost is a major factor.

3. THIN-FILM (AMORPHOUS) PANELS

- **MANUFACTURE:** Made by depositing a very thin layer of photovoltaic material (like amorphous silicon) onto a backing (like glass or flexible plastic).
- **APPEARANCE:** Very thin, often black, and can be **flexible**.
- **PROS:**
 - **Very Flexible and Lightweight:** Can be integrated into building materials (e.g., roofing tiles) or used for portable applications.
 - **Best Heat Tolerance:** Performs well in very hot climates.
- **CONS:**
 - **Lowest Efficiency:** Typically 10-13%.
 - **Requires a huge amount of space** for a useful amount of power.
 - Degrades faster than crystalline panels.
- **APPLICATION:** Niche uses: portable solar chargers, boats, RVs, building-integrated PV.

SOLAR PANEL RATINGS (THE STICKER)

On the back of every solar panel is a "spec sheet" sticker. This is the panel's "birth certificate" and is critical for matching it to an inverter. The key ratings are:

- **P_{max} (Maximum Power):** This is the "name" of the panel (e.g., 300W). It's the maximum power the panel will produce under Standard Test Conditions (STC).
- **V_{mpp} (Voltage at Maximum Power):** The "sweet spot" voltage where the panel produces its maximum power.
- **I_{mpp} (Current at Maximum Power):** The "sweet spot" current (Amps) where the panel produces its maximum power.
 - $(V_{mpp} \times I_{mpp} = P_{max})$
- **V_{oc} (Open Circuit Voltage):** The *maximum voltage* the panel can produce (this happens when it's sunny but the panel is *not connected* to anything).
- **I_{sc} (Short Circuit Current):** The *maximum current* the panel can produce (this happens if the positive and negative wires are touched together).

EVALUATION

1. What is the difference between a "solar cell," a "solar module," and a "solar array"?
2. Describe the appearance of a Monocrystalline panel. What is its main advantage?
3. Describe the appearance of a Polycrystalline panel. What is its main advantage?
4. What does "Pmax" mean on a panel's rating sticker?
5. What does "Voc" mean, and why is it an important safety rating?

ASSIGNMENT

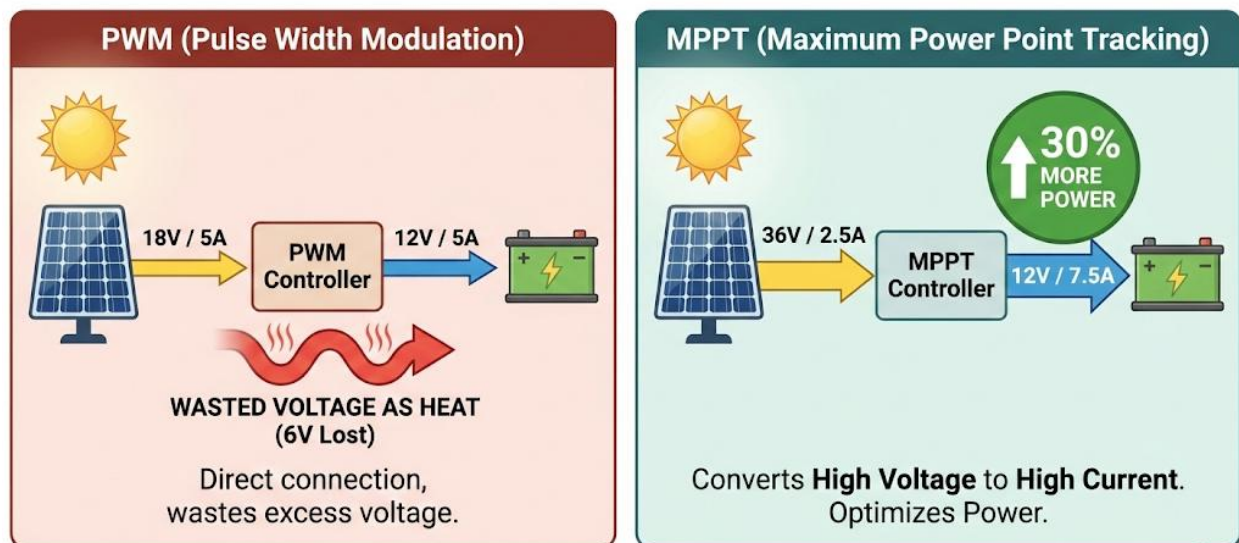
1. Create a comparison table for Monocrystalline, Polycrystalline, and Thin-Film panels. Use the criteria: **Efficiency, Cost, Appearance, and Best Application.**
2. Write a 200-word essay answering: "If you have a very small roof in a hot city like Lagos, which type of panel would you choose, and why?"
3. Go online and find a "spec sheet" for any solar panel (e.g., search "350W solar panel spec sheet"). Write down the values for its Pmax, Vmpp, Impp, Voc, and Isc.
4. Define the following terms: Semiconductor, Wafer, P-N Junction, Amorphous, Array.
5. If a panel has a Vmpp of 36V and an Impp of 9A, what is its Pmax (Maximum Power)? (Show your calculation).

WEEK 8: COMPONENTS OF PV SYSTEMS II (CHARGE CONTROLLERS)

FUNCTION OF A CHARGE CONTROLLER

The **Charge Controller** (or "Charge Regulator") is the "**brain**" of an off-grid or hybrid solar system. It is a small box that sits between the **Solar Panels** and the **Battery Bank**.

PWM vs. MPPT Charge Controllers: Efficiency Comparison



Summary: Demonstrates why MPPT is more efficient, especially with high-voltage panels.

Its job is to be a "traffic cop" for the electricity. It has two primary, critical functions:

1. PREVENT BATTERY OVERCHARGING:

- On a sunny day, the panels can produce a lot of power. If the battery becomes full, this excess power would "cook" the battery, causing it to overheat, bubble (gassing), and be destroyed.
- The charge controller senses the battery's voltage. When it gets full, the controller **stops or reduces** the flow of power from the panels.

2. PREVENT BATTERY OVER-DISCHARGING:

- At night, the appliances (loads) draw power *from* the battery. If they are allowed to drain the battery completely (to 0%), the battery will be permanently damaged (sulfation).
- The charge controller senses the battery's voltage. If it gets too low, the controller will **disconnect the loads** to protect the battery from a "deep discharge."⁷⁸

TYPES OF CHARGE CONTROLLERS

There are two main technologies for charge controllers.

1. PWM (PULSE WIDTH MODULATION)

- **TECHNOLOGY:** This is the older, simpler, and cheaper technology.
- **HOW IT WORKS:** It works like a simple, rapid switch.
 - When the battery is low, the switch is "ON" 100% of the time, allowing all panel power to flow in.
 - As the battery gets fuller, the controller "pulses" the switch (on-off-on-off) very quickly. This "**Pulse Width**" (the amount of "on" time) gets shorter and shorter, "tapering" the charge.
- **VOLTAGE MATCHING:** A PWM controller **requires** the solar panel's voltage to be *very close* to the battery's voltage.
 - e.g., You must use a "12V" panel (which is ~18V V_{mpp}) with a "12V" battery.
- **PROS:**
 - **Cheap** and reliable.
 - Good for small, simple systems (e.g., a single panel, a solar street light).
- **CONS:**
 - **Inefficient.** It's a "power-waster." If you connect a 250W/30V panel to a 12V battery, the PWM controller will "throw away" the extra voltage, only using ~12V. You will lose over 50% of your panel's power.

2. MPPT (MAXIMUM POWER POINT TRACKING)

- **TECHNOLOGY:** This is the modern, "smart," and more expensive technology.
- **HOW IT WORKS:** An MPPT controller is a sophisticated **DC-to-DC converter**.
 - It *finds* the panel's "Maximum Power Point" (the V_{mpp} and I_{mpp} "sweet spot").
 - It then "**converts**" the panel's high-voltage/low-current power into the battery's low-voltage/high-current power.
 - **Example:** A 250W/30V/8.3A panel is connected to a 12V battery.
 - The MPPT controller takes the **30 Volts** from the panel.

- It "converts" it down to the battery's charging voltage (e.g., ~14V).
- To keep the *power* the same ($\text{Power} = V \times A$), it *boosts* the **current (Amps)**.
- Power in: $30V \times 8.3A = 250W$
- Power out: $14V \times \sim 17.8A = 250W$
- It has "harvested" **17.8 Amps** for the battery, while a PWM would have only given **8.3 Amps**.
- **PROS:**
 - **Highly Efficient (up to 99%):** Can "harvest" 15-30% more power from your array than a PWM.
 - **Flexible:** Allows you to use high-voltage (cheaper, large) panels with low-voltage batteries (e.g., 24V or 48V panels on a 12V battery).
- **CONS:**
 - **More Expensive.**
- **APPLICATION:** Essential for *any* serious or large solar system.

EVALUATION

1. What are the two primary, critical functions of a charge controller?
2. What happens to a battery if it is (a) overcharged or (b) over-discharged?
3. What does "PWM" stand for, and how does it control the charge?
4. What does "MPPT" stand for, and why is it more efficient than a PWM?
5. If you connect a 30V solar panel to a 12V battery, which type of charge controller *must* you use to get all the panel's power?

ASSIGNMENT

1. Draw a simple block diagram showing a panel, a charge controller, a battery, and a DC load (like a 12V light bulb).
2. Write a 250-word essay explaining the MPPT "power conversion" example from the notes, showing how voltage is traded for current.
3. Go online and find prices for a "30A PWM" charge controller and a "30A MPPT" charge controller. Compare their costs.

4. You are designing a small solar system for a single security light. The system has one 50W (18V) panel and one 12V battery. Which charge controller (PWM or MPPT) would you use, and why?
5. Define the following terms: Overcharge, Deep Discharge, Tapering, DC-to-DC Converter, Voltage Matching.

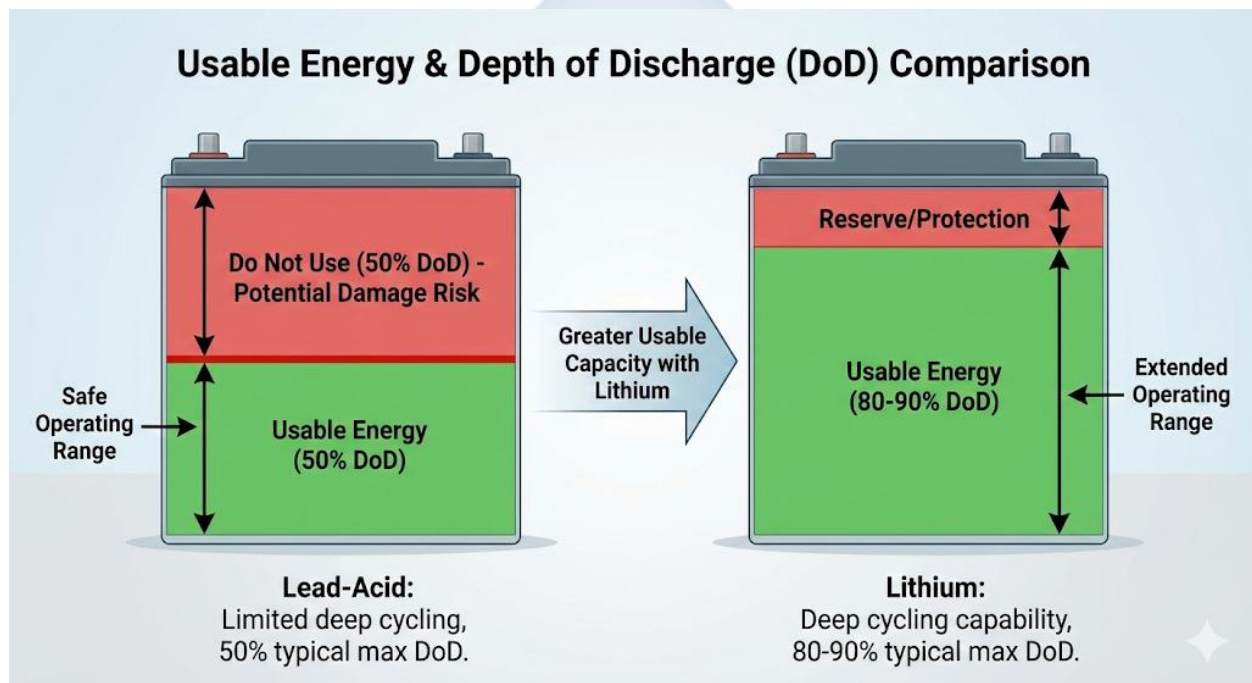


WEEK 9: COMPONENTS OF PV SYSTEMS III (BATTERIES)

FUNCTION OF A BATTERY BANK

The **Battery** (or "Battery Bank" when multiple are connected) is the **storage tank** of a solar system.

- **FUNCTION:** Its sole purpose is to **store the DC energy** produced by the panels during the day, so that this energy can be used at **night** or during **cloudy/rainy days**.
- **ROLE:** Batteries are *essential* for **Off-Grid** and **Hybrid** systems. They are *not* used in On-Grid systems.



TYPES OF BATTERIES (BATTERY CHEMISTRY)

There are two main families of batteries used in solar.

1. LEAD-ACID BATTERIES

- **TECHNOLOGY:** This is the oldest, most traditional battery technology. It uses plates of **Lead (Pb)** submerged in an **electrolyte (Sulphuric Acid)**.
- **SUB-TYPES:**
 - **Flooded (or FLA):** The "classic" battery. The lead plates are submerged in *liquid* acid.

- *Pros:* Cheapest option.
- *Cons:* Requires regular **maintenance** (checking and topping up with *distilled water*), must be kept upright, releases flammable hydrogen gas (needs ventilation).
- **Sealed (AGM or Gel):** These are "maintenance-free."
- **AGM (Absorbent Glass Mat):** The acid is absorbed in a fiberglass mat.
- **Gel:** The acid is mixed into a thick gel.
- *Pros:* Spill-proof, no maintenance.
- *Cons:* More expensive than flooded, can be damaged by overcharging.

2. LITHIUM-ION (LFP/LiFePO4) BATTERIES

- **TECHNOLOGY:** This is the modern, new technology. The most common type for solar is **Lithium Iron Phosphate (LiFePO4)**.
- **PROS:**
 - **High "Depth of Discharge" (DoD):** (See below).
 - **Long Lifespan:** Can last 10-15 years (5,000+ cycles).
 - **Lightweight and Compact:** Much smaller for the same capacity.
 - **High Efficiency:** Can charge and discharge very quickly.
 - **Zero Maintenance.**
- **CONS:**
 - **Very High Cost:** The main barrier.

BATTERY SIZING AND RATINGS

- **VOLTAGE (V):** Batteries come in standard voltages, (e.g., 12V). You must connect batteries in **series** (to add voltage) or **parallel** (to add capacity) to match your inverter (e.g., four 12V batteries in series = a 48V bank).
- **CAPACITY (AMP-HOURS, Ah):** This is the "size" of the battery. A 100Ah battery can *in theory* supply 10 Amps for 10 hours, or 1 Amp for 100 hours.
- **DEPTH OF DISCHARGE (DoD):** This is the *most important* concept for battery health.

- **Definition:** DoD is the percentage of the battery's total capacity that you can *safely* use without damaging it.
- **Lead-Acid:** Has a poor DoD, typically **50%**. This means if you have a 200Ah lead-acid battery, you must *only use 100Ah* before you recharge it.
- **Lithium-ion:** Has an excellent DoD, typically **80-90%**. This means on a 200Ah lithium battery, you can *safely use 160-180Ah*.
- **Conclusion:** A 100Ah Lithium battery has almost the same *usable* energy as a 200Ah Lead-Acid battery.

BATTERY CARE AND MAINTENANCE

1. **VENTILATION:** Flooded lead-acid batteries *must* be stored in a well-ventilated area to allow flammable hydrogen gas to escape.
2. **TEMPERATURE:** Batteries hate high heat.⁸⁸ Heat (e.g., a hot generator room) will drastically shorten a battery's life.
3. **CLEANLINESS:** Keep the battery tops and terminals clean and free of corrosion (white powder).
4. **MAINTENANCE (FLOODED ONLY):** Regularly check the electrolyte (water) level and top up *only* with **distilled water**.
5. **OBEY THE DoD:** The charge controller must be set to the correct DoD (e.g., 50% for lead-acid) to prevent over-discharging and ensure a long life.

EVALUATION

1. What is the primary function of a battery bank in a solar system?
2. What are the two main types (chemistries) of batteries used in solar?
3. What is the main *maintenance* requirement for a "Flooded" lead-acid battery?
4. Define "Depth of Discharge (DoD)."
5. Why does a 100Ah Lithium-ion battery provide almost the same *usable power* as a 200Ah Lead-Acid battery?

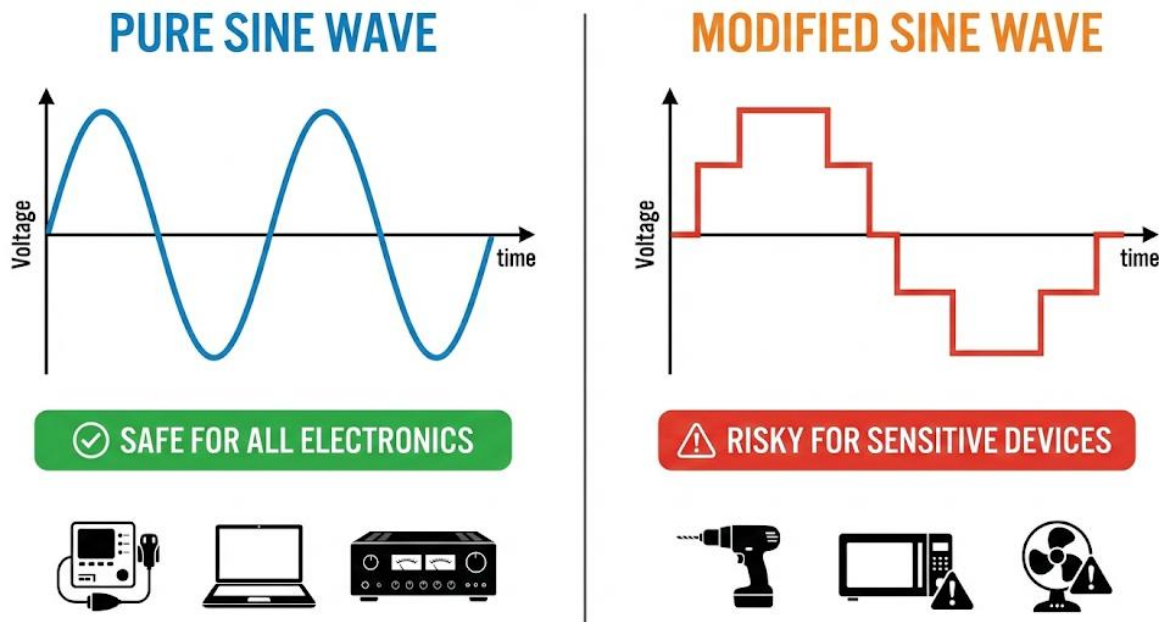
ASSIGNMENT

1. Create a comparison table for **Flooded Lead-Acid** vs. **Lithium-ion (LiFePO₄)**. Use the criteria: **Cost, Lifespan, DoD, Maintenance, and Weight**.
2. You have a 12V, 200Ah Flooded Lead-Acid battery. If your charge controller is set to a 50% DoD, how many *usable* Amp-hours (Ah) do you have?
3. What is "series" battery connection and what is "parallel" battery connection? Draw a simple diagram showing how you would connect *two* 12V batteries to make a 24V bank.
4. Research and list 5 safety rules for handling and storing batteries. (Hint: ventilation, no sparks, lift correctly, etc.).
5. Define the following terms: Electrolyte, Sulfation, Distilled Water, AGM, LiFePO₄.



WEEK 10: COMPONENTS OF PV SYSTEMS IV (INVERTERS)

FUNCTION OF AN INVERTER



The **Inverter** is the "**heart**" of the PV system. It is arguably the most complex and important component.

- **PRIMARY FUNCTION:** Its main job is to **convert** the low-voltage **Direct Current (DC)** electricity (from the panels or batteries) into high-voltage **Alternating Current (AC)** electricity.
- **WHY?**
 - Solar panels and batteries produce and store **DC**.
 - All our standard household appliances (lights, fans, TVs, fridges, sockets) run on **AC** (e.g., 220-240V AC).
 - The inverter "inverts" the power from one form to the other, making it usable.

TYPES OF INVERTERS (BY WAVEFORM)

The "quality" of the AC electricity an inverter produces is called its **waveform**. There are two types.

1. MODIFIED SINE WAVE INVERTERS

- **WAVEFORM:** Produces a "blocky" or "choppy" wave. It is a rough approximation of AC power.
- **PROS:**
 - **Very Cheap.**
 - Simple technology.
- **CONS:**
 - **Can damage sensitive electronics:** (e.g., laptops, computers, modern TVs, medical equipment).⁹⁵
 - **Inefficient with "Inductive Loads":** (e.g., motors, fans, refrigerators, pumps). It makes them run hotter, less efficiently, and can shorten their lifespan.
 - Often produces a "buzzing" sound from audio equipment.
- **APPLICATION:** Only suitable for *very simple* loads like basic light bulbs or old tube TVs. **Generally not recommended for a modern home.**

2. PURE SINE WAVE INVERTERS

- **WAVEFORM:** Produces a "smooth," clean, curved wave. This waveform is **identical to (or better than)** the electricity you get from the utility grid (NEPA).
- **PROS:**
 - **Safe for all appliances:** It can run any electronic device, from a laptop to a sensitive medical device, without risk of damage.
 - **Efficient:** Runs motors and inductive loads (like fridges and ACs) quietly and at their correct efficiency.
- **CONS:**
 - **More Expensive** than modified sine wave.
- **APPLICATION:** This is the **standard** and **required** type for any serious solar installation (home, business, hospital).

TYPES OF INVERTERS (BY SYSTEM FUNCTION)

- **OFF-GRID INVERTER:**

- A simple inverter that *only* knows how to take DC power from a **battery** and turn it into AC for the house. It cannot connect to the grid.

- **GRID-TIED INVERTER:**

- A "smart" inverter that takes DC power *directly from the panels* and converts it to AC that is *synchronized* with the grid. It does *not* work with batteries.

- **HYBRID INVERTER:**

- The "all-in-one" solution. This single box contains:
 1. An **MPPT Charge Controller** (to charge batteries from panels).
 2. A **Battery Inverter** (to convert battery DC to AC).
 3. A **Grid-Tied Inverter** (to sync with the grid).
 4. A **Battery Charger** (to charge batteries from the grid/generator).
- This is the most common "inverter" sold in Nigeria for modern solar systems.

INVERTER RATINGS

- **POWER (WATTS / KILOWATTS / KVA):**

- This is the inverter's "size" (e.g., 1.5kW, 3.5kVA, 5kW). It tells you the *maximum amount of load* (appliances) it can power *at one time*.

- **VOLTAGE (DC):**

- This is the **DC input voltage** (e.g., 12V, 24V, 48V).
- This rating is **critical**. The inverter's DC voltage **MUST** match the voltage of your battery bank (e.g., a 48V inverter needs a 48V battery bank).

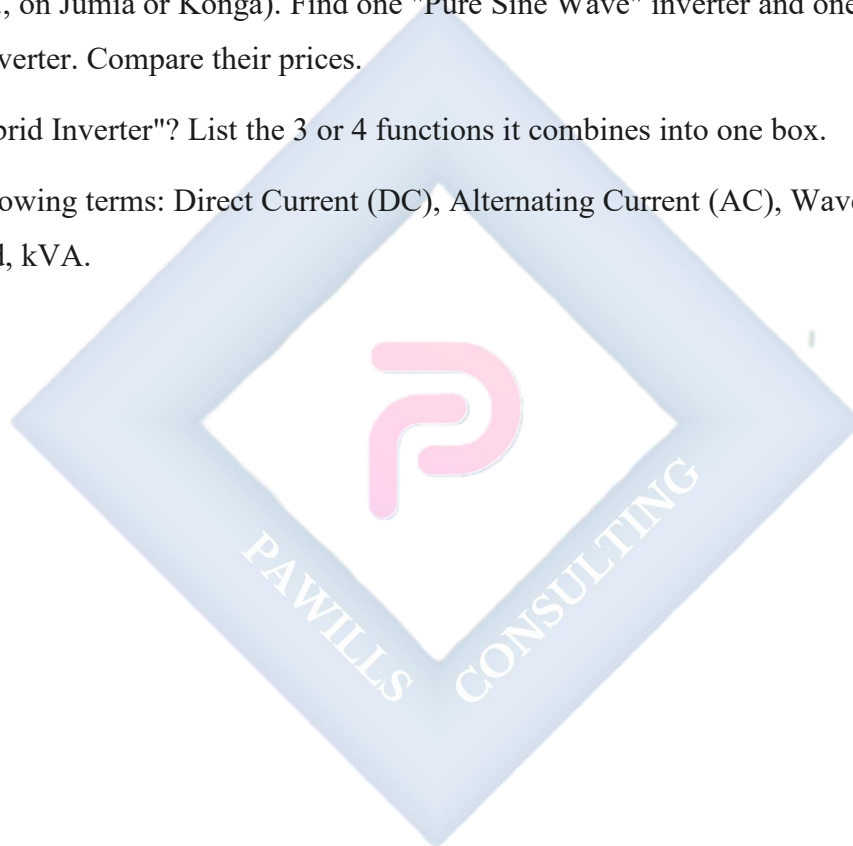
EVALUATION

1. What is the primary function of an inverter? Why is this necessary?
2. What are the two main types of inverter *waveforms*?
3. Why is a "Modified Sine Wave" inverter dangerous for a refrigerator or a laptop?
4. What type of waveform is produced by a "Pure Sine Wave" inverter?

5. What are the two main ratings of an inverter, and what do they mean?

ASSIGNMENT

1. Draw a simple diagram showing the flow of power: Battery (DC) → Inverter → Light bulb (AC).
2. Write a 250-word essay explaining to a new client why they *must* buy a **Pure Sine Wave** inverter for their home, even though it is more expensive than a Modified Sine Wave inverter.
3. Go online (e.g., on Jumia or Konga). Find one "Pure Sine Wave" inverter and one "Modified Sine Wave" inverter. Compare their prices.
4. What is a "Hybrid Inverter"? List the 3 or 4 functions it combines into one box.
5. Define the following terms: Direct Current (DC), Alternating Current (AC), Waveform, Inductive Load, kVA.



WEEK 11-13: REVISION & EXAMS

REVISION OF ALL FIRST TERM TOPICS

This period is dedicated to a comprehensive review of all components and concepts covered this term, in preparation for the final examination.

- **Week 1-3 (Foundations):**

- Renewable vs. Non-Renewable.
- The Photovoltaic Effect (Photon $\rightarrow$ Electron $\rightarrow$ DC Current).
- Key Terms: **Irradiance** (Power, W/m^2), **Insolation** (Energy, $\text{kWh/m}^2/\text{day}$), **PSH** (Sizing value), **Efficiency** (%).

- **Week 4 (The Systems):**

- **Off-Grid:** Needs batteries, no grid.
- **On-Grid:** Needs grid, no batteries. (Fails when grid fails).
- **Hybrid:** Needs grid *and* batteries. (Best for Nigeria).

- **Week 6-10 (The 4 Main Components):**

1. **SOLAR PANELS:**

- Types: **Monocrystalline** (Black, high efficiency), **Polycrystalline** (Blue, cheaper).
- Ratings: P_{max} , V_{oc} , I_{sc} .

2. **CHARGE CONTROLLERS:**

- Function: Protects the battery (no overcharge, no over-discharge).
- Types: **PWM** (cheaper, inefficient, needs voltage match), **MPPT** (efficient, converts $V \rightarrow A$).

3. **BATTERIES:**

- Function: Stores DC energy.
- Types: **Lead-Acid** (cheaper, 50% DoD, needs maintenance), **Lithium-ion** (expensive, 80-90% DoD, no maintenance).
- Key Term: **Depth of Discharge (DoD)**.

4. **INVERTERS:**

- Function: Converts **DC $\rightarrow$ AC**.

- Types: **Modified Sine** (cheap, dangerous for electronics), **Pure Sine** (essential for all modern appliances).
- Ratings: Power (kW/kVA), Voltage (12V, 24V, 48V - must match battery).

FINAL EXAM

The final examination will test all aspects of this term's work, including definitions, comparisons, simple calculations, and scenario-based problem-solving.



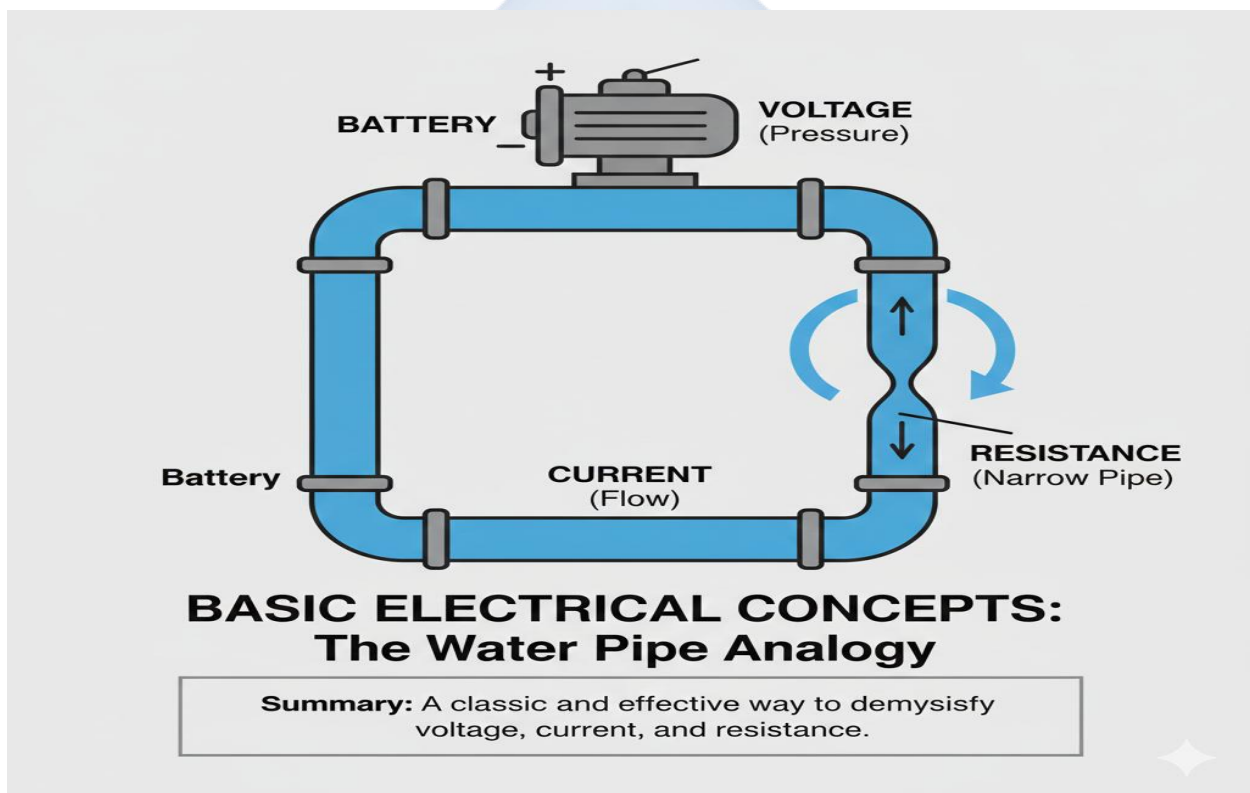
SS1 SOLAR PHOTOVOLTAIC INSTALLATION & MAINTENANCE LESSON NOTES

SECOND TERM

WEEK 1: BASIC ELECTRICAL CONCEPTS

INTRODUCTION TO ELECTRICITY

Before we can install, maintain, or even connect a solar panel, we must understand the "invisible" force it produces: electricity. Electricity is the flow of electrons. To understand it, we will use a common and effective analogy: **visualizing electricity as water flowing through a pipe.**



CURRENT (I)

- **DEFINITION:** Current is the **flow rate** of electric charge (electrons) past a point in a circuit. It is the "amount" of electricity that is moving.
- **ANALOGY (WATER):** Current is the **volume of water flowing** through the pipe (e.g., 5 Liters per second). A high current is a "gushing" flow, while a low current is a "trickle."

- **UNIT OF MEASUREMENT: Amperes (A),** or "Amps."
- **IN A SOLAR CONTEXT:** A 300W panel might produce 8 Amps of current in full sun. This is the "volume" of electricity it is generating.

VOLTAGE (V)

- **DEFINITION: Voltage** (also called "Potential Difference") is the **electrical pressure** or "force" that pushes the current through the circuit. It is the *potential* for energy to move.
- **ANALOGY (WATER):** Voltage is the **water pressure** in the pipe. A high-pressure pipe (high voltage) can "push" a lot of water (current) very hard. A low-pressure pipe (low voltage) has a weak "push."
- **UNIT OF MEASUREMENT: Volts (V).**
- **IN A SOLAR CONTEXT:** A "12V" panel produces an electrical pressure of about 18-20 Volts. This "pressure" is what pushes the current to the battery.

RESISTANCE (R)

- **DEFINITION: Resistance** is the **opposition** to the flow of current. It is anything that "resists" or slows down the electricity. All materials have some resistance (except superconductors).
- **ANALOGY (WATER):** Resistance is a **narrowing or crimp in the pipe**. It's the "friction" inside the pipe. A thin pipe has high resistance, while a thick, wide pipe has low resistance.
- **UNIT OF MEASUREMENT: Ohms (Ω).**
- **IN A SOLAR CONTEXT:** The wires (cables) used in an installation have resistance. A wire that is **too thin** or **too long** will have high resistance. This is bad, as it wastes energy (lost as heat) and reduces the voltage.

POWER (P)

- **DEFINITION: Power** is the **rate at which electrical energy is used** or generated. It is the *total work* being done.

- **ANALOGY (WATER):** Power is the **total force** of the water coming out of the hose. It is the *combination* of the pressure (Voltage) and the flow rate (Current). A high-pressure, high-flow hose has high power.
- **UNIT OF MEASUREMENT: Watts (W).** (1000 Watts = 1 Kilowatt, kW).
- **IN A SOLAR CONTEXT:** This is the "size" of a panel (e.g., 300 Watts) or an appliance (e.g., a 60-Watt fan).

OHM'S LAW AND THE POWER LAW

These are the "golden rules" of electricity.

1. **OHM'S LAW:** This law describes the relationship between Voltage, Current, and Resistance.
 - **Formula:** $V = I \times R$ (Voltage = Current $\times$ Resistance)
 - **Explanation:** The "pressure" (V) needed is equal to the "flow" (I) multiplied by the "friction" (R).
 - *To find Current:* $I = V / R$
 - *To find Resistance:* $R = V / I$
2. **THE POWER LAW (OR WATT'S LAW):** This describes the relationship between Power, Voltage, and Current.
 - **Formula:** $P = V \times I$ (Power = Voltage $\times$ Current)
 - **Explanation:** The total "power" (P) is the "pressure" (V) multiplied by the "flow" (I).
 - *To find Current:* $I = P / V$
 - *To find Voltage:* $V = P / I$

EXAMPLE CALCULATION:

A 120-Watt (P) television runs on a 12-Volt (V) solar system. How much current (I) does it draw?

- Use Power Law: $I = P / V$
- $I = 120W / 12V = 10A$
- This means you need a wire that can safely handle 10 Amps of current.

EVALUATION

1. In your own words, define Voltage, Current, and Resistance using the water pipe analogy.
2. What is the standard unit of measurement for (a) Current, (b) Voltage, (c) Resistance, and (d) Power?
3. State the formula for Ohm's Law.
4. State the formula for the Power Law.
5. A solar panel is connected to a resistor. The Voltage is 12V and the Current is 3A. What is the Resistance (in Ohms)? (Show your calculation).

ASSIGNMENT

1. Write a 200-word essay explaining the "water in a pipe" analogy for V, I, and R.
2. A 12-Volt solar system is powering a load that has 4 Ohms of resistance. How much current (in Amps) is flowing? (Show your calculation).
3. A 50-Watt (P) light bulb is running on a 12-Volt (V) battery. How much current (I) does it draw? (Show your calculation).
4. Research and define the difference between **Direct Current (DC)** and **Alternating Current (AC)**. Which one do solar panels produce?
5. List three common household appliances and find their "Power" rating in Watts (W) (it is usually on a sticker on the back).

WEEK 2: TOOLS & INSTRUMENTS IN PV INSTALLATION

INTRODUCTION

A solar installer is a technical professional. Like a doctor or a mechanic, an installer is only as good as their tools. Using the wrong tool is unprofessional, slow, and dangerous.

We divide the tools into two groups: **Diagnostic/Measuring Instruments** (the "brains") and **Hand/Power Tools** (the "muscle").



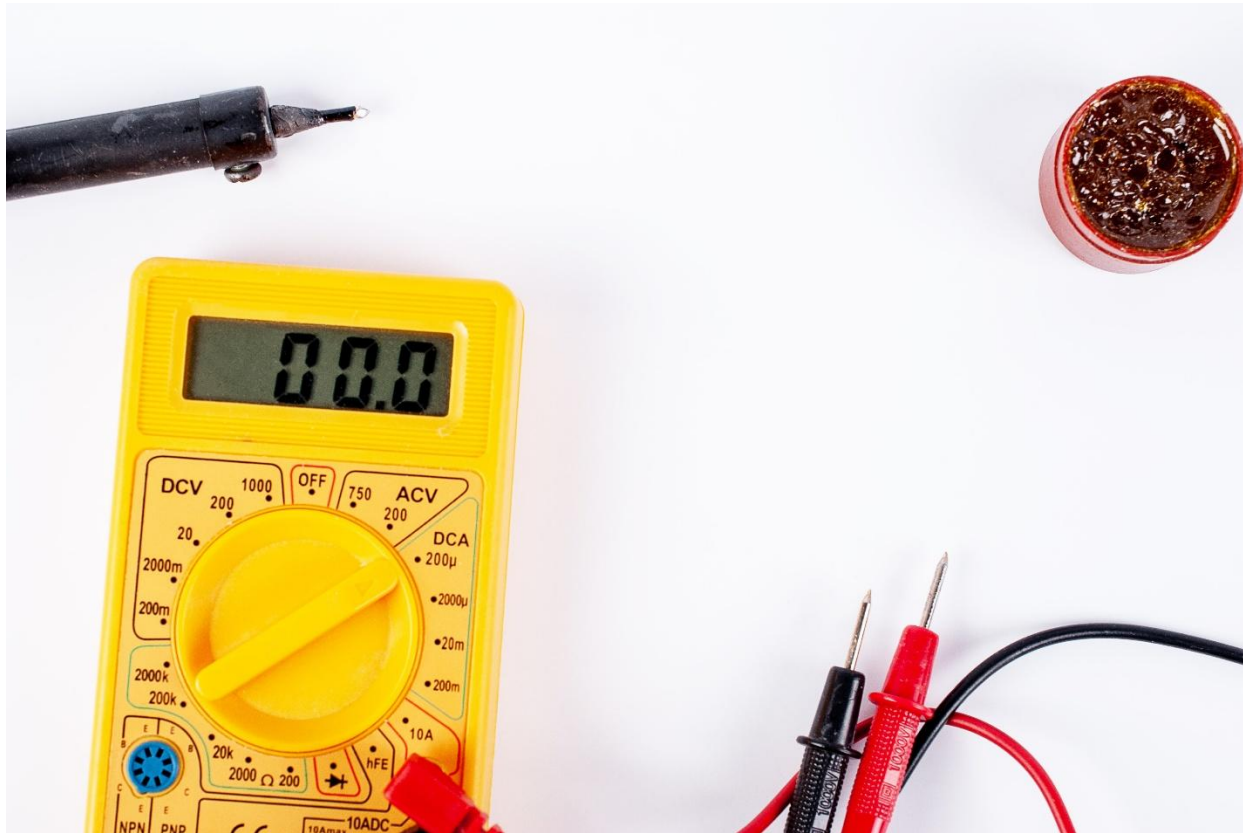
DIAGNOSTIC & MEASURING INSTRUMENTS

These tools are used to "see" electricity. They are essential for installation, troubleshooting, and maintenance.

1. DIGITAL MULTIMETER (DMM):

- **Description:** This is the installer's most important diagnostic tool. It is a handheld device with a digital screen and two probes (one red, one black).
- **Core Functions:**

- **Voltmeter (Measures Voltage):** Used to check the voltage of panels (Voc) and batteries. **How to Use:** You connect the probes in **PARALLEL** with the component (touch red to positive, black to negative).



- **Ammeter (Measures Current):** Used to check the current of a panel (Isc). **How to Use:** You must connect the meter in **SERIES** (like a link in a chain). This requires *breaking* the circuit, which can be dangerous.
 - **Ohmmeter (Measures Resistance):** Used to check for broken wires (continuity) or test fuses. **How to Use:** Must be used *only* when the system is **OFF** (de-energized).
- **Safety:** Always set the dial to the *correct* measurement (DC Volts, AC Volts, Amps, Ohms) *before* you connect it.

2. CLAMP METER:

- **Description:** A DMM, but with a large "jaw" or "clamp" at the top.
- **Primary Function:** To measure **CURRENT (AMPS)** easily and safely.

- **How to Use:** Instead of breaking the circuit, you simply **clamp the jaw around a single wire**. The meter uses the wire's magnetic field to measure the current flowing through it.
 - **Advantage:** This is the *preferred* way to measure high current, as it is much safer and faster than using a traditional ammeter.
3. **SOLAR TESTER (SOLAR IRRADIANCE METER):**
- **Description:** A small handheld device that measures the "brightness" of the sun.
 - **Function:** It measures **Solar Irradiance** in **Watts per square meter (W/m^2)**.
 - **Why it's used:** If a customer says their system is not working well, this is the first tool you use. You measure the sunlight.
 - *If the meter reads $900 W/m^2$ (strong sun) but the panel is producing low power, you know the panel is faulty.*
 - *If the meter reads $200 W/m^2$ (heavy clouds), you know the panel is fine—it's just not getting any "fuel."*

HAND TOOLS & SAFETY GEAR (PPE)

These are the tools used for the physical installation.

- **WIRE STRIPPERS & CUTTERS:**
 - **Wire Cutter:** For cutting cables to the correct length.
 - **Wire Stripper:** A specialized tool to remove the outer plastic insulation from a wire *without* cutting the copper inside.
- **MC4 CRIMPING TOOL:**
 - **MC4 Connectors** are the standard waterproof, locking connectors used on all modern solar panels.
 - This **specialized tool** is used to "crimp" (attach) the metal pin of an MC4 connector to the end of a solar cable. *Using regular pliers will result in a bad, unsafe connection that will overheat.*
- **SPANNERS (WRENCHES) & SOCKET SET:**
 - Used to tighten the nuts and bolts that hold the **mounting structure** (the racks) onto the roof or ground.
- **SCREWDRIVERS (PHILLIPS & FLAT-HEAD):**

- Used for attaching components like charge controllers, inverters, and circuit breakers to the wall.
- **PERSONAL PROTECTIVE EQUIPMENT (PPE):**
 - **Safety Goggles:** To protect eyes from drilling debris or battery acid.
 - **Insulated Gloves:** To protect from electric shock.
 - **Hard Hat:** When working on a busy construction site or under other installers.
 - **Safety Boots:** To protect feet and provide grip on a roof.

EVALUATION

1. What are the three main functions of a Digital Multimeter (DMM)?
2. How do you connect a voltmeter to measure a battery's voltage (Series or Parallel)?
3. Why is a Clamp Meter generally safer than a DMM for measuring current (Amps)?
4. What does a Solar Irradiance Meter measure, and what is its unit?
5. What is an MC4 Crimping Tool used for? Why can't you use regular pliers?

ASSIGNMENT

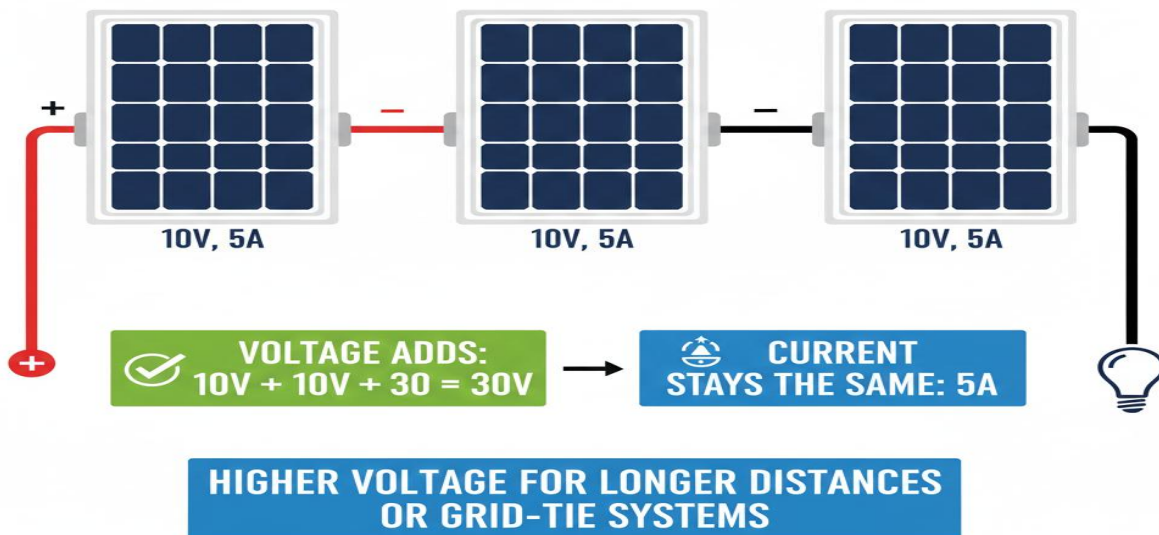
1. Draw a simple circuit diagram showing how to correctly connect a **Voltmeter** (in Parallel) to measure the voltage of a battery.
2. Draw a simple circuit diagram showing how to correctly connect an **Ammeter** (in Series) to measure the current flowing from a battery to a light bulb.
3. Draw a diagram showing how a **Clamp Meter** is used (clamped *around* one wire).
4. Research "MC4 Connectors." Draw a simple sketch of a male and female connector and explain why they are used on solar panels.
5. Create a "Safety First" poster, listing 5 essential PPE items for a solar installer and what each item protects against.

WEEK 3: SOLAR PANEL CONNECTIONS I (SERIES)

INTRODUCTION

A single solar panel (module) does not produce enough voltage or current to power a house. To get useful power, we must connect multiple panels together in a "solar array." The two ways to connect them are **Series** and **Parallel**.

SOLAR PANEL CONNECTIONS: SERIES

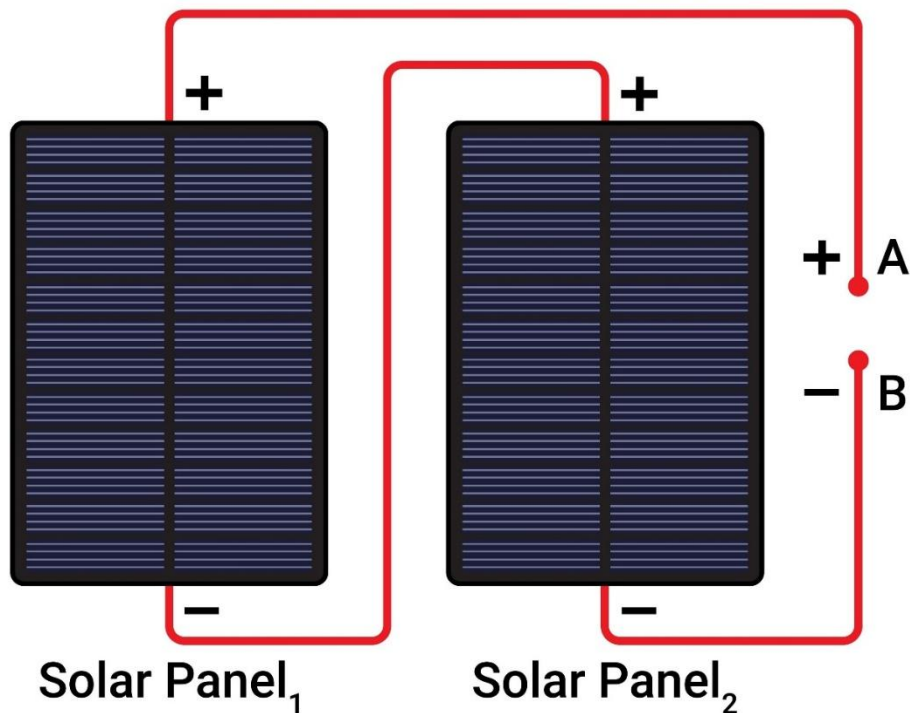


This week, we focus on **Series connections**.

WHAT IS A SERIES CONNECTION?

A **series connection** is when panels are connected "head-to-tail" in a single chain. The **positive (+) terminal** of the first panel is connected to the **negative (-) terminal** of the second panel. The positive (+) of the second to the negative (-) of the third, and so on.

2 Solar Panels In Series



- **ANALOGY:** Think of a **TV remote**. You connect the batteries in a chain, (+) to (-), to make one long "string" of batteries.

THE LAW OF SERIES CONNECTIONS: VOLTAGE EFFECTS

This is the "golden rule" for series connections. When you connect components in **SERIES**:

VOLTAGE (V) ADDS, but CURRENT (I) STAYS THE SAME.

CALCULATION & EXAMPLE

Let's assume we have **3 identical solar panels** with the following ratings:

- **1 Panel:**

- Voltage (V_{mpp}) = 20 Volts
- Current (I_{mpp}) = 5 Amps
- Power = $20V \times 5A = 100W$

Now, let's connect all 3 panels in **SERIES**:

1. **TOTAL VOLTAGE (V):** The voltages add up.
 - $V_{total} = V_{panel1} + V_{panel2} + V_{panel3}$
 - $V_{total} = 20V + 20V + 20V = 60V$
2. **TOTAL CURRENT (I):** The current stays the same as the *lowest* panel in the string.
 - $I_{total} = I_{panel1} = I_{panel2} = I_{panel3}$
 - $I_{total} = 5A$
3. **TOTAL POWER (P):** The total power is the new total voltage $\times$ the total current.
 - $P_{total} = V_{total} \times I_{total}$
 - $P_{total} = 60V \times 5A = 300W$
 - (This makes sense: $3 \times 100W$ panels = $300W$ total).

WHY USE A SERIES CONNECTION?

1. **To Increase System Voltage:** This is the *main* reason. Many inverters and charge controllers *require* a high input voltage (e.g., 24V, 48V, or even 150V for an MPPT). You use a series connection to "stack" the voltage of your panels to meet this requirement.
2. **To Use Thinner Cables:** Remember $P = V \times I$. If you increase the Voltage, the Current (Amps) can stay low for the same amount of power. Low current allows you to use **thinner, cheaper cables** over long distances without significant "voltage drop" or fire risk.

THE PROBLEM WITH SERIES: SHADING

- **THE "CHRISTMAS LIGHT EFFECT":** This is the *biggest disadvantage* of a series connection.

- **Analogy:** The current (Amps) in a series string is like water flowing through one single pipe. If you put a "crimp" (a blockage) *anywhere* in that pipe, the flow of the *entire* pipe is reduced.
- **In a solar array:** If *one panel* is covered by a shadow (from a tree, a chimney, or bird droppings), its current drops to near zero.
- **Result:** Because the current *must* be the same for the whole string, the current of **ALL** the panels in that string drops to the low level of the one shaded panel. You can lose **90%** of your power just from **10%** shading.

EVALUATION

1. Define a "Series Connection" in your own words.
2. What is the golden rule for Voltage and Current in a series connection?
3. You have two 12-Volt, 8-Amp panels. What is the total Voltage and total Current when you connect them in series? (Show your calculation).
4. What is the main *advantage* (purpose) of connecting panels in series?
5. What is the "Christmas Light Effect," and how does shading impact a series string?

ASSIGNMENT

1. Draw a clear, labeled wiring diagram showing **four** 100W (20V, 5A) panels connected in series.
2. For the diagram you just drew, calculate the final:
 - a) Total Voltage
 - b) Total Current
 - c) Total Power
3. Look at the batteries in a large torchlight (flashlight) or a TV remote. Are they connected in series or parallel? How do you know?
4. Research "Bypass Diodes" in a solar panel. Write a 150-word report on what they are and how they help solve the shading problem in a series string.

5. Why is it better to design a 48V system (using series connections) rather than a 12V system if the wire from the panels to the house is very long (e.g., 50 meters)? (Hint: Think about resistance and voltage drop from Week 1 & 8).



WEEK 4: SOLAR PANEL CONNECTIONS II (PARALLEL)

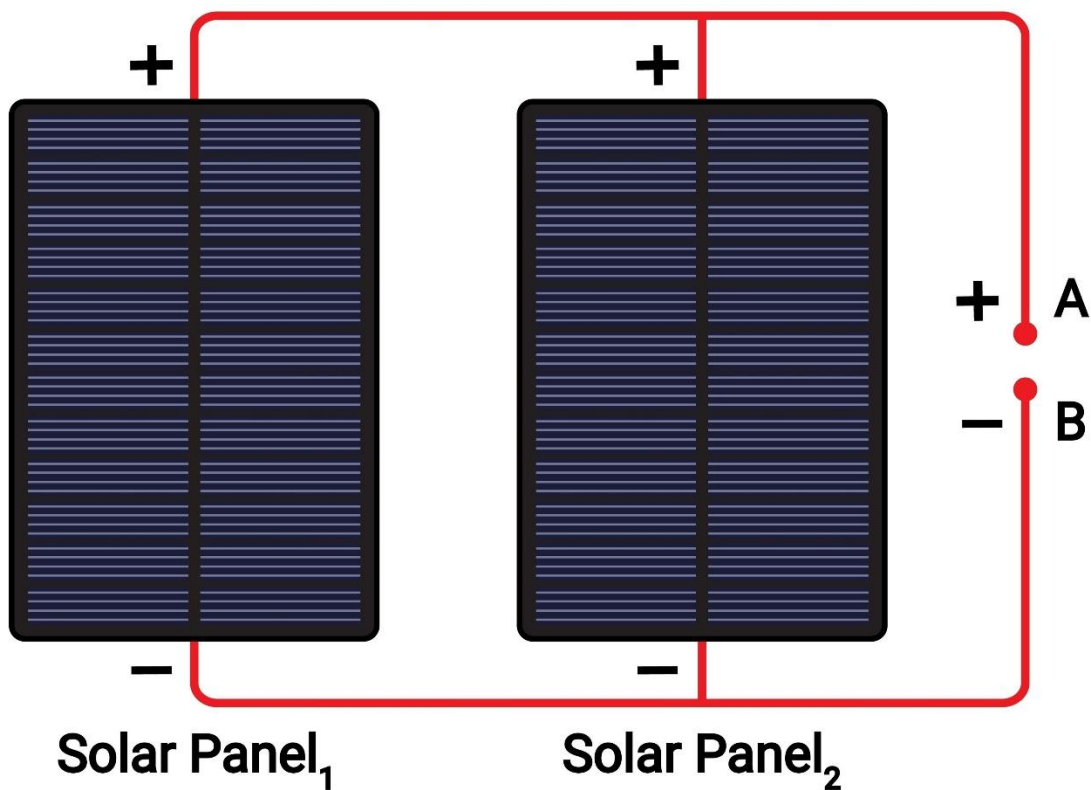
INTRODUCTION

The second way to connect panels is in parallel. This method has the *opposite* effect of a series connection and is used for different reasons.

WHAT IS A PARALLEL CONNECTION?

A **parallel connection** is when all panels are connected "Positive to Positive" and "Negative to Negative." All the positive terminals are joined together, and all the negative terminals are joined together.

2 Solar Panels in Parallel



- **ANALOGY:** Think of rungs on a **ladder**. Each panel is a separate path for the current to flow.
- **How it's done:** You use **MC4 "branch connectors"** (Y-connectors) to combine all the positive wires into one, and all the negative wires into one.

THE LAW OF PARALLEL CONNECTIONS: CURRENT EFFECTS

This is the "golden rule" for parallel connections. When you connect components in PARALLEL:

CURRENT (I) ADDS, but VOLTAGE (V) STAYS THE SAME.

CALCULATION & EXAMPLE

Let's use the *same 3 identical solar panels* from last week:

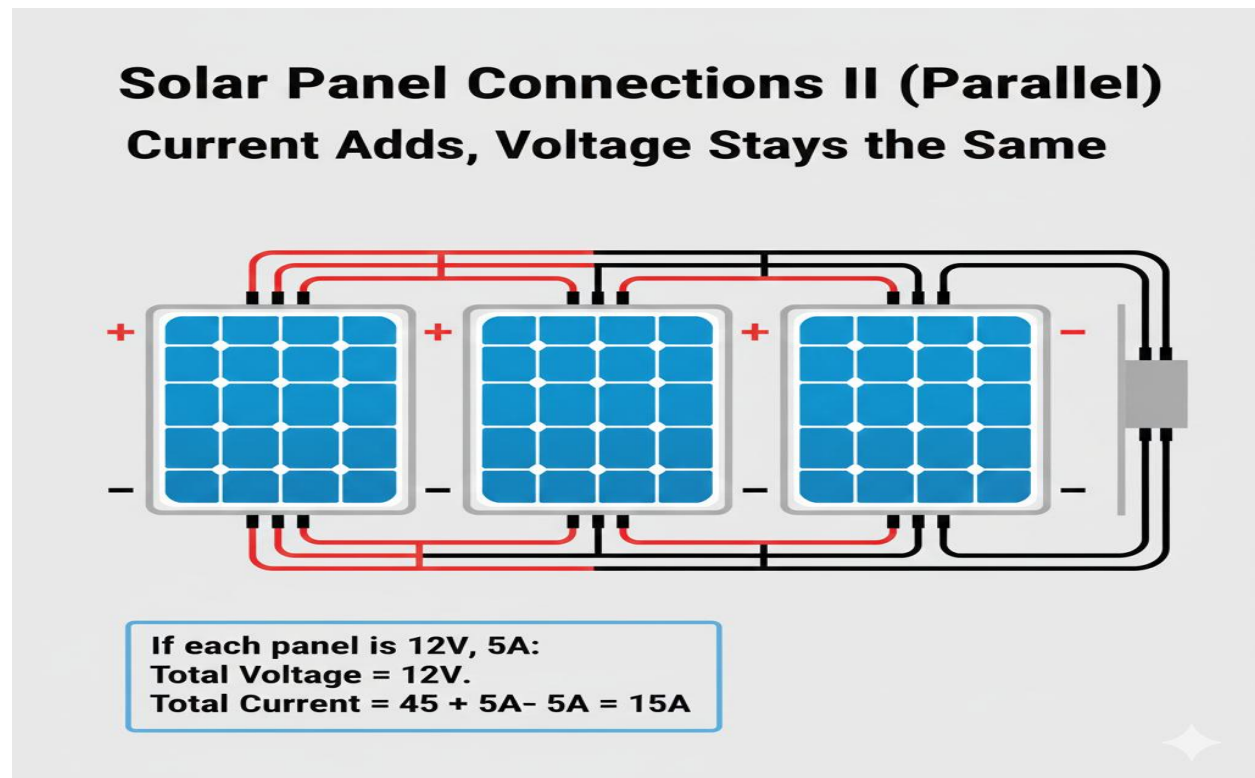
- **1 Panel:**
 - Voltage (V_{mpp}) = 20 Volts
 - Current (I_{mpp}) = 5 Amps
 - Power = $20V \times 5A = 100W$

Now, let's connect all 3 panels in **PARALLEL**:

1. **TOTAL VOLTAGE (V):** The voltage stays the same as the *lowest* panel in the group.
 - $V_{total} = V_{panel1} = V_{panel2} = V_{panel3}$
 - $V_{total} = 20V$
2. **TOTAL CURRENT (I):** The currents add up.
 - $I_{total} = I_{panel1} + I_{panel2} + I_{panel3}$
 - $I_{total} = 5A + 5A + 5A = 15A$
3. **TOTAL POWER (P):** The total power is the new total voltage $\times$ the total current.
 - $P_{total} = V_{total} \times I_{total}$
 - $P_{total} = 20V \times 15A = 300W$
 - (The power is the same as the series connection, just achieved in a different way).

WHY USE A PARALLEL CONNECTION?

1. **To Increase Current (Amperage):** This is the main reason. Some charge controllers (especially older PWM types) are limited by voltage (e.g., a "12V" controller) but can handle high current. You use parallel to *increase* the charging current *without* exceeding the controller's voltage limit.
2. **To Improve Shade Tolerance:** This is the *biggest advantage* of a parallel connection.



SHADING IN A PARALLEL CONNECTION

- **Analogy:** If a series string is *one pipe*, a parallel array is *three separate pipes* flowing into one big bucket.
- **Result:** If one panel (one pipe) is shaded (crimped), **only that single panel's current is lost**. The other two unshaded panels (pipes) continue to deliver their *full* 5 Amps each.
- **Example:** If one of our 3 parallel panels is shaded:
 - Shaded Panel = 0A
 - Panel 2 = 5A
 - Panel 3 = 5A

- Total Current = $0 + 5 + 5 = 10\text{A}$
- The system *still produces 2/3 of its power*. A series string would have dropped to near 0A.

SERIES-PARALLEL CONNECTIONS (THE REAL WORLD)

In large systems, you combine *both* methods.

- *Example:* For a 48V, 30A system, you might:
 1. Create a "string" of **2 panels in series** ($20\text{V}+20\text{V} = 40\text{V}$, 5A).
 2. Create *another* string of **2 panels in series** ($20\text{V}+20\text{V} = 40\text{V}$, 5A).
 3. Connect those two *strings* in **parallel**.
 4. **Final Output:** Voltage stays at 40V. Current adds ($5\text{A}+5\text{A} = 10\text{A}$).

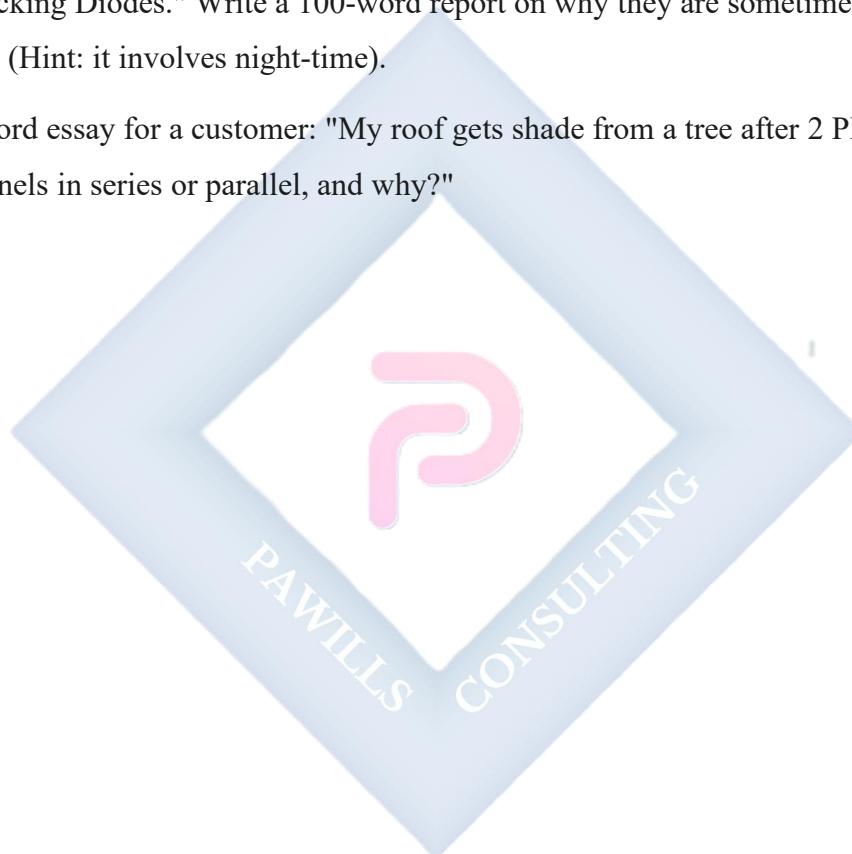
EVALUATION

1. Define a "Parallel Connection" in your own words.
2. What is the golden rule for Voltage and Current in a parallel connection?
3. You have two 12-Volt, 8-Amp panels. What is the total Voltage and total Current when you connect them in parallel? (Show your calculation).
4. What is the main *advantage* of a parallel connection, especially regarding shading?
5. In a parallel array of 4 panels, if one panel is 100% shaded, how much of the *total power* is lost? (e.g., 25%, 50%, 100%?)

ASSIGNMENT

1. Draw a clear, labeled wiring diagram showing **four** 100W (20V, 5A) panels connected in parallel. (Hint: You will need branch connectors).
2. For the diagram you just drew, calculate the final:
 - a) Total Voltage

- b) Total Current
 - c) Total Power
3. **Challenge:** You have **six** 100W (20V, 5A) panels and need to make a **40V** system.
- a) How would you connect them (series, parallel, or series-parallel)?
 - b) Draw the diagram for your connection.
 - c) What is the final Voltage, Current, and Power of your 6-panel array?
4. Research "Blocking Diodes." Write a 100-word report on why they are sometimes needed in parallel strings (Hint: it involves night-time).
5. Write a 200-word essay for a customer: "My roof gets shade from a tree after 2 PM. Should I connect my panels in series or parallel, and why?"



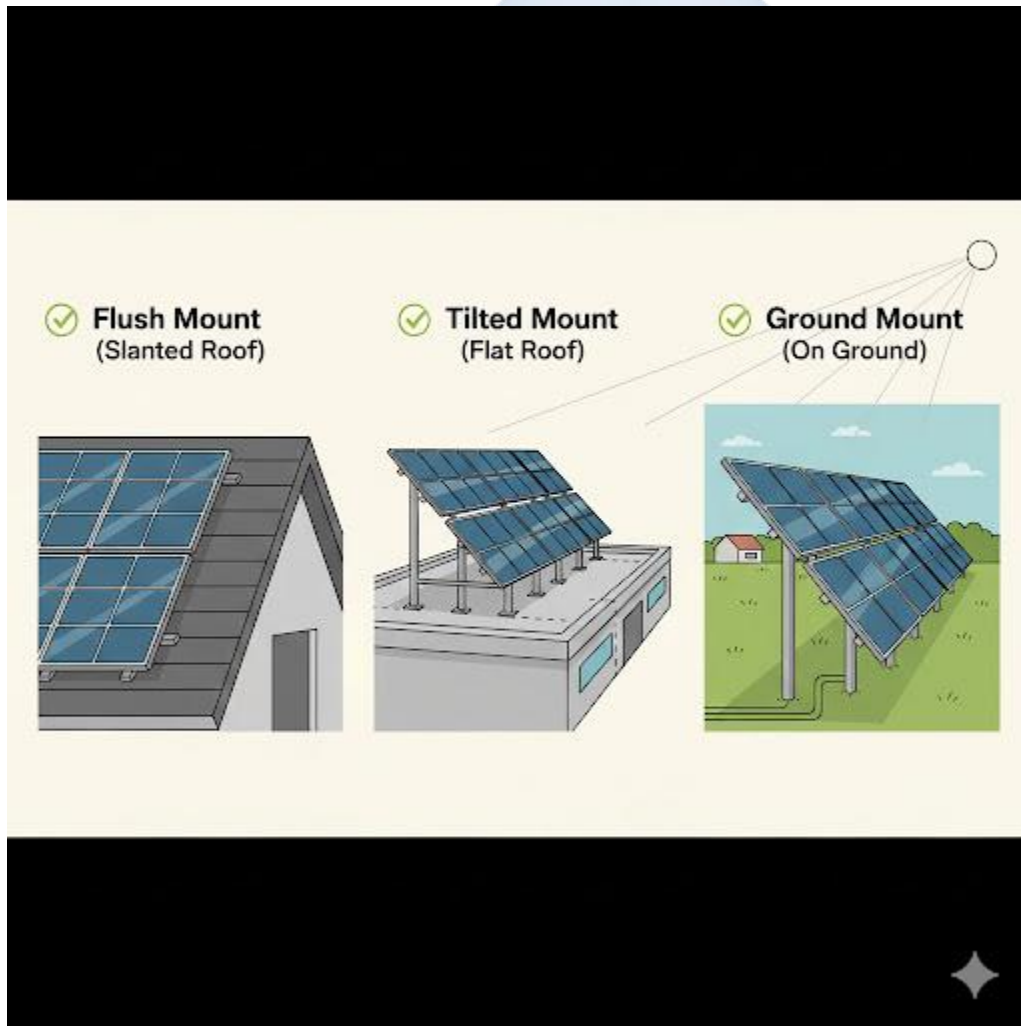
WEEK 6: SOLAR PANEL MOUNTING STRUCTURES

INTRODUCTION

A solar panel is useless if it's lying flat on the ground or in the shade. The **mounting structure** (or "racking system") is the framework that securely fastens the solar panels to a surface.

A *good* mounting system does three things:

1. **Secures** the panels safely against wind and weather.
2. **Orients** the panels at the correct *azimuth* (direction).
3. **Tilts** the panels at the correct *angle* to maximize sun exposure.



SITE ASSESSMENT: THE FIRST STEP

Before you can choose a mount, you must perform a **Site Assessment**. This is a professional survey to find the *best* location for the panels.

- **SHADE ANALYSIS:** This is the *most important* part. You must check for any objects that will cast a shadow on the panels *at any time of day, all year long*.
 - *Common problems:* Trees, other buildings, chimneys, satellite dishes, water tanks.
- **ROOF ASSESSMENT:**
 - *Integrity:* Is the roof strong enough to hold the panels and the installers?
 - *Material:* Is it a zinc (metal) roof, tile, or concrete? This determines the type of mounting bolts needed.
 - *Space:* Is there enough *unshaded* space?
- **SUN PATH ANALYSIS:** Identifying the path of the sun (East → South → West) to determine the correct orientation.

ORIENTATION (AZIMUTH)

- **DEFINITION:** Orientation (or "Azimuth") is the **compass direction** the panels are facing (North, South, East, West).
- **THE RULE:** The sun rises in the East and sets in the West, but it "tracks" across the *southern* sky.
- **FOR NIGERIA (Northern Hemisphere):** To get the maximum, most consistent sun exposure all day, solar panels *must* face **TRUE SOUTH**.
 - Facing **East** is "okay" (you get strong morning sun, weak afternoon sun).
 - Facing **West** is "okay" (you get weak morning, strong afternoon).
 - Facing **North** is **unacceptable** and will result in very poor production.

TILT ANGLE

- **DEFINITION:** The **Tilt Angle** is the angle of the panel relative to the horizontal ground.
 - 0° = flat on the ground.

- 90° = vertical on a wall.
- **THE RULE:** The sun is high in the sky in the summer and low in the sky in the winter. The *ideal* tilt is one that is perpendicular to the sun for most of the year.
- **THE "RULE OF THUMB":** For a fixed, year-round system, the ideal tilt angle is **equal to the latitude** of the location.
- **EXAMPLE IN NIGERIA:**
 - Lagos Latitude: $\sim 6.5^\circ$ N $\rightarrow$ Ideal Tilt = **6.5°**
 - Abuja Latitude: $\sim 9^\circ$ N $\rightarrow$ Ideal Tilt = **9°**
 - Kano Latitude: $\sim 12^\circ$ N $\rightarrow$ Ideal Tilt = **12°**
 - *Note:* Since most roofs are steeper than this, installers often mount panels "flush" (flat) to the roof, which is "good enough." On a flat concrete roof, however, they *must* use a tilted mount.

TYPES OF MOUNTING STRUCTURES

1. ROOF MOUNTS (MOST COMMON):

- **Flush Mount:** Panels are mounted parallel (flat) to the roof, a few inches above it to allow airflow. This is the cheapest and simplest option.
- **Tilted Mount (Tripod):** Used on *flat* concrete roofs. A metal frame (often an A-frame) is used to *create* the correct tilt angle (e.g., 10°). These must be weighed down with blocks.

2. GROUND MOUNTS:

- A large rack is built on the ground.
- **Pros:** Can be set at the *perfect* orientation and tilt. Easy to access for cleaning.
- **Cons:** Uses valuable land, requires a concrete foundation, and has a higher security risk (theft).

3. POLE MOUNTS:

- Panels are mounted on top of a single, tall pole, like a streetlight.
- **Pros:** Gets the panels high above ground-level shade.
- **Cons:** Expensive, difficult to install.

4. SOLAR TRACKERS (ADVANCED):

- A motorized, "smart" mount that *follows the sun* across the sky, just like a sunflower.
- **Pros:** Can increase energy production by 25-40%.

- **Cons:** Extremely expensive, complex, and has moving parts that can break.

EVALUATION

1. What are the three main purposes of a solar panel mounting structure?
2. What is "Shade Analysis," and why is it the most important part of a site assessment?
3. What is the correct **Orientation (Azimuth)** for solar panels in Nigeria?
4. What is the "Rule of Thumb" for determining the ideal **Tilt Angle**?
5. What is the difference between a "Flush Mount" and a "Tilted Mount"?

ASSIGNMENT

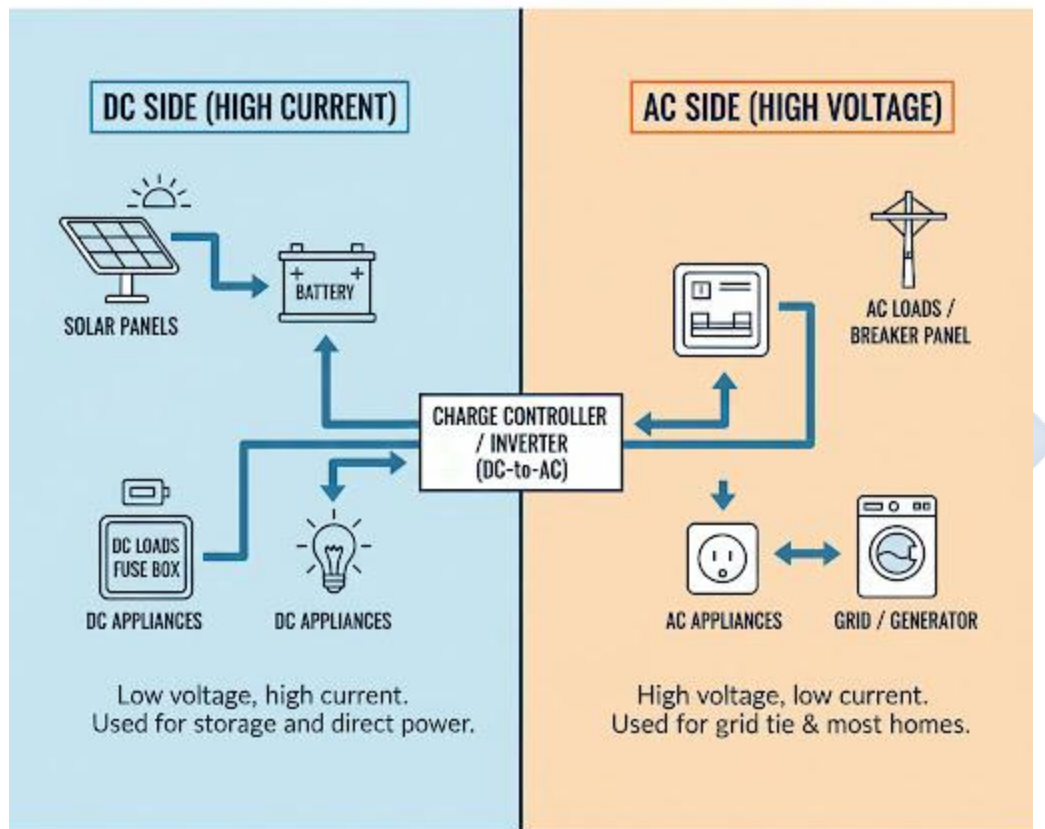
1. What is the latitude of your current city? (You can search this online). Based on this, what is the *ideal* tilt angle for solar panels in your location?
2. Using a compass or a compass app on a phone, stand outside and find **True South**. Is your own roof (or the roof of your school) facing in a good direction to install solar panels?
3. Draw a simple diagram of a house with a *flat* roof. Show how a **Tilted Mount** would be used to angle the panel.
4. Create a "Pros and Cons" table comparing **Roof Mounting** vs. **Ground Mounting**.
5. Research "Single-Axis Solar Tracker" vs. "Dual-Axis Solar Tracker." Write a 150-word report on the difference.

WEEK 8: WIRING & CABLING

INTRODUCTION

A solar system is an electrical system. The "veins" that carry the power from one component to another are the **wires and cables**. Choosing the wrong cable is one of the most common and *most dangerous* mistakes in an installation.

HYBRID ELECTRICAL SYSTEM: DC VS. AC SIDES



Two distinct electrical environments requiring different components and wiring for safety and efficiency.

DC & AC CABLING

A hybrid or off-grid system has two distinct "sides" of wiring:

1. THE DC (DIRECT CURRENT) SIDE:

- **Location:** This is all the wiring *before* the inverter.
- *Path 1:* Solar Panels → Charge Controller
- *Path 2:* Charge Controller → Battery Bank
- *Path 3:* Battery Bank → Inverter
- **Characteristics:** This side often involves *low voltage* (e.g., 12V, 24V, 48V) but *very high current* (Amps).

2. THE AC (ALTERNATING CURRENT) SIDE:

- **Location:** This is all the wiring *after* the inverter.
- *Path:* Inverter → Distribution Board (Fuse Box) → Appliances (Lights, Sockets, Fans).
- **Characteristics:** This side involves *high voltage* (e.g., 220V) but *low current*.

CABLE SIZING (THE CRITICAL CONCEPT)

You cannot just use "any wire." The "size" (thickness) of the cable is critical.

- **ANALOGY:** Using a wire that is too thin is like trying to drink a thick yam porridge through a tiny straw. It's very difficult, and you won't get much.
- **THE PROBLEM: RESISTANCE.** A thin wire has *high resistance*. When you force a high current (Amps) through a high-resistance wire, two bad things happen:
 1. **VOLTAGE DROP:** The "pressure" (Voltage) is lost as friction. By the time the electricity gets to the end of the wire, the voltage has "dropped." A 12V battery might only deliver 10V to the inverter, causing it to shut down.
 2. **HEAT:** The friction is converted into **heat**. The wire gets hot, which wastes precious solar energy. In a worst-case scenario, the wire will *melt* its insulation and **cause a fire**.

THE RULES OF SIZING:

- **Rule 1:** The **higher the CURRENT (Amps)**, the **THICKER** the cable must be.
- **Rule 2:** The **longer the DISTANCE** (e.g., from panels to inverter), the **THICKER** the cable must be.

WHY 48V IS BETTER THAN 12V:

- Recall $P = V \times I$. To get 1200 Watts of power:
 - At 12V: $1200W / 12V = 100$ Amps (This is a *massive* current, requiring a very thick, expensive cable).
 - At 48V: $1200W / 48V = 25$ Amps (This is a manageable current that can use a much thinner, cheaper cable).
- This is why all large solar systems use higher voltages (48V or more).
- **UNITS:** Cable size is measured in **mm²** (square millimeters of the copper's cross-sectional area) or **AWG** (American Wire Gauge).

SAFETY & CABLE TYPES

- **SOLAR CABLE (PV CABLE):**
 - The cable used *outdoors* to connect the panels must be special.
 - It must be **UV-resistant** (so the sun doesn't crack its insulation) and waterproof.
- **BATTERY CABLE:**
 - The cables connecting the battery to the inverter.
 - These are usually the *thickest* cables in the system (to handle the high current).
- **COLOR CODING:** Using the correct color is a critical safety rule.
 - **DC Side:** **Red** for Positive (+), **Black** for Negative (-).
 - **AC Side:** (In Nigeria) **Red/Brown** (Live), **Blue** (Neutral), **Green/Yellow** (Earth).
- **CONNECTIONS:** All connections must be *tight*. A loose connection creates high resistance and heat, and is a major fire hazard.

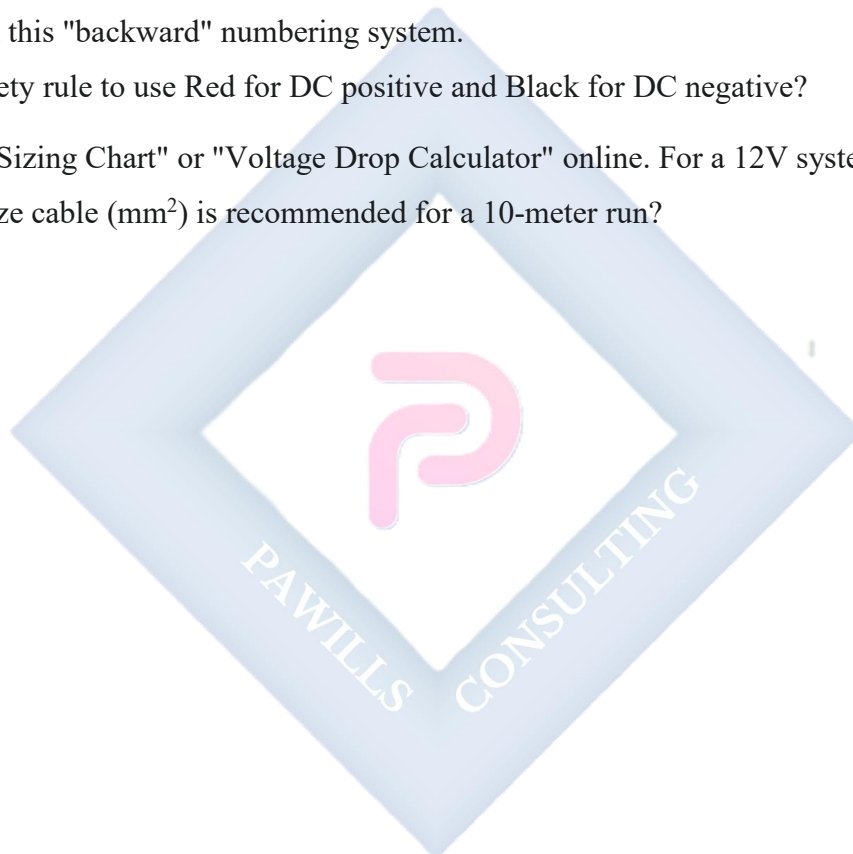
EVALUATION

1. What is the "DC side" of a solar installation? What is the "AC side"?
2. What are the two *dangers* of using a wire that is too thin for the current it is carrying?
3. What is "Voltage Drop"?
4. Why does a high-power 12V system require a much thicker cable than a 48V system for the *same amount of power*?

5. What special property must the cable used for outdoor solar panels have?

ASSIGNMENT

1. Draw a full block diagram of a hybrid solar system and *label* the "DC Cabling" sections and the "AC Cabling" sections.
2. A 12V system runs a load of 60 Amps. A 48V system runs a load of 15 Amps. Which system will suffer from *more* voltage drop and heat loss, and why?
3. Research "American Wire Gauge (AWG)." Is a 10 AWG cable *thicker* or *thinner* than a 4 AWG cable? Explain this "backward" numbering system.
4. Why is it a safety rule to use Red for DC positive and Black for DC negative?
5. Find a "Cable Sizing Chart" or "Voltage Drop Calculator" online. For a 12V system drawing 20 Amps, what size cable (mm²) is recommended for a 10-meter run?



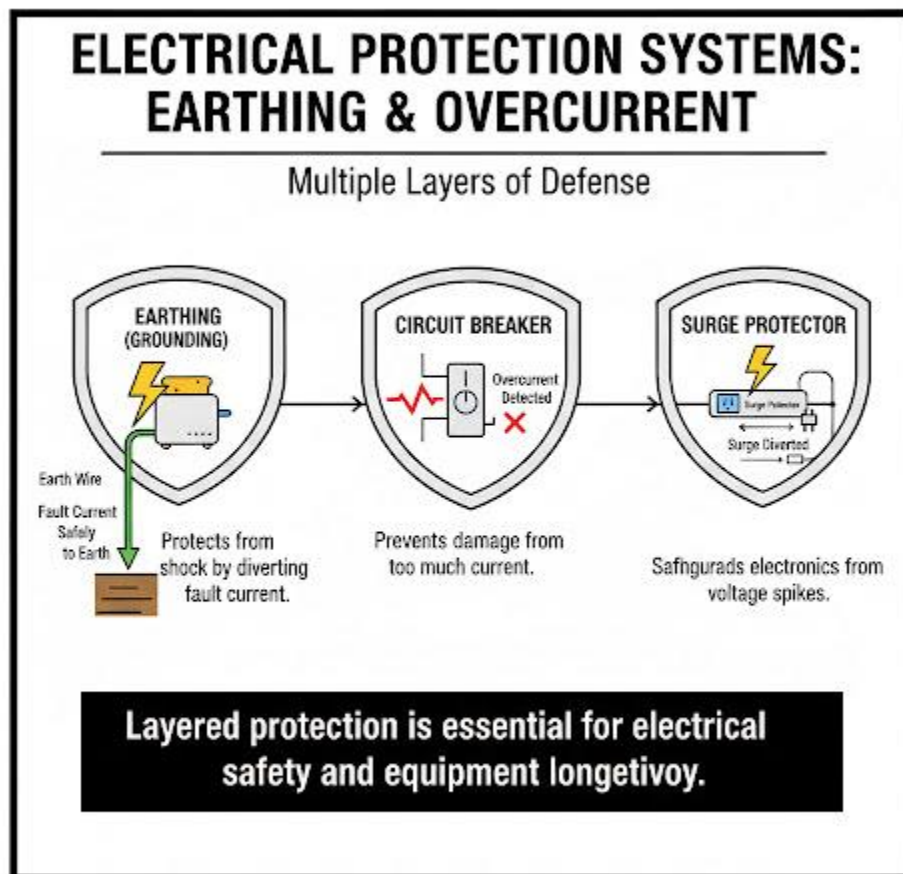
WEEK 9: EARTHING & PROTECTION

INTRODUCTION

A solar installation is a power plant. It can be dangerous. **Protection devices** are the safety features that prevent **fire, equipment damage, and electric shock**. A system built without protection is a "time bomb."

There are two types of protection:

1. **People Protection:** Earthing (Grounding).
2. **Equipment Protection:** Breakers, Fuses, and Surge Protectors.



EARTHING (GROUNDING)

- **DEFINITION: Earthing** (or "Grounding") is the process of creating a safe, low-resistance path for electricity to flow to the **Earth** (the ground) in case of a fault.
- **COMPONENTS:** This is done by driving a long copper rod (a **Grounding Rod** or **Earth Electrode**) deep into the soil and connecting wires from it to the system.
- **PURPOSE (SAFETY FOR PEOPLE):**
 - All *non-current-carrying* metal parts of the system must be earthed.
 - *Examples:* The aluminum **frames of the solar panels**, the metal **mounting rack**, and the metal **case of the inverter**.
 - **How it works:** Imagine a live wire inside the inverter breaks and touches the metal case. The case is now *live* (at 220V).
 - *Without Earthing:* If you touch the inverter, *you* become the path to the ground. You will get a severe electric shock.
 - *With Earthing:* The earth wire provides a "superhighway" for the fault current to flow safely to the ground, tripping the circuit breaker and preventing you from being shocked.

OVERCURRENT PROTECTION (CIRCUIT BREAKERS & FUSES)

- **DEFINITION: Overcurrent** is a situation where *too much current (Amps)* flows through a wire. This is caused by a "short circuit" or an overloaded appliance.
- **THE DANGER:** As we learned in Week 8, high current = high heat. Overcurrent will *melt* the wire's insulation and **cause a fire**.

CIRCUIT BREAKERS (AC & DC):

- **Function: A re-usable safety switch.** It constantly monitors the current. If the current exceeds its rating (e.g., 20A), it automatically "trips" (shuts off), breaking the circuit and stopping the fire.
- **Location:** Breakers must be installed in *all* key locations:
 - Between Panels and Charge Controller.
 - Between Charge Controller and Battery.
 - Between Battery and Inverter.

- Between Inverter and AC Loads (in the Distribution Board).

FUSES:

- **Function:** A **one-time use** device. It has a thin wire inside that is designed to *melt* and break the circuit if the current is too high.
- **Disadvantage:** Must be manually replaced after it "blows."
- **Location:** Often found inside inverters and in special "battery fuses."

SURGE PROTECTION (SURGE PROTECTORS, SPDs)

- **DEFINITION:** A "Surge" (or "Spike") is a *sudden, massive increase in voltage* that lasts for a microsecond.
- **SOURCE:** The most common source is **LIGHTNING**. A lightning strike *near* the house can send a 100,000-Volt spike down the wires.
- **THE DANGER:** This high-voltage spike will instantly **destroy** sensitive electronics, especially the **inverter** and **charge controller**.
- **SURGE PROTECTION DEVICE (SPD):**
 - **Function:** A device that acts like a "trap." It sits on the wire and waits. During normal operation, it does nothing.
 - **How it works:** When it "sees" a high-voltage spike (like lightning), it instantly "opens" a path and **diverts** that dangerous energy safely to the **earth (ground)**. It "sacrifices" itself to save the expensive equipment.
 - **Location:** SPDs are installed on both the DC side (near the panels) and the AC side (in the distribution board).

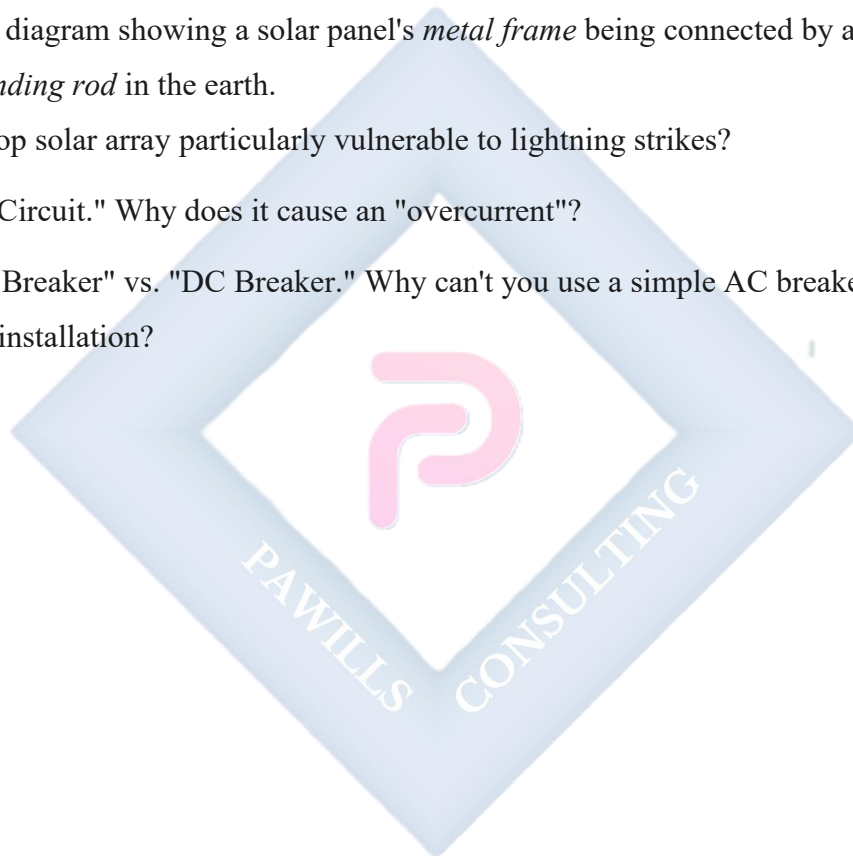
EVALUATION

1. What is the primary purpose of **Earthing** (Grounding)? (Protection for People or Equipment?)
2. What is the primary purpose of a **Circuit Breaker**? (Protection for People or Equipment/Wires?)
3. What is the main *difference* between a Circuit Breaker and a Fuse?
4. What does a **Surge Protection Device (SPD)** protect equipment from?

5. List three metal parts of a PV system that must be connected to the earth rod.

ASSIGNMENT

1. Go to your home's main distribution board (fuse box). Open it (with adult supervision). Identify the main switch and the individual circuit breakers. Write down the "Amp" rating (e.g., 15A, 30A) of three different breakers.
2. Draw a simple diagram showing a solar panel's *metal frame* being connected by a green/yellow wire to a *grounding rod* in the earth.
3. Why is a rooftop solar array particularly vulnerable to lightning strikes?
4. Define "Short Circuit." Why does it cause an "overcurrent"?
5. Research "AC Breaker" vs. "DC Breaker." Why can't you use a simple AC breaker on the DC side of a solar installation?



WEEK 10: PRACTICAL: SMALL SOLAR SETUP

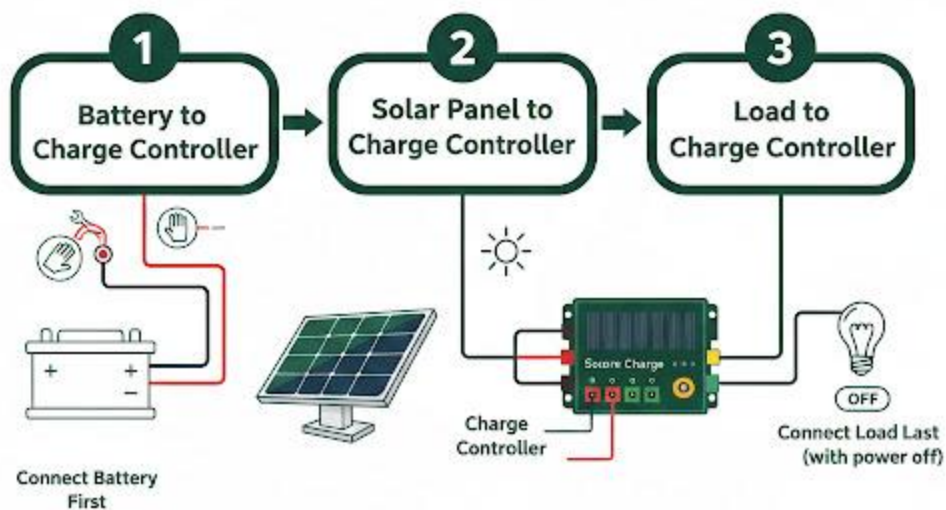
TOPIC: ASSEMBLING A 12V DC LIGHTING SYSTEM

OBJECTIVE

To apply all the theoretical knowledge from this term and the last term to build a simple, functional, **off-grid DC-only system**. This practical demonstrates the fundamental connection sequence and principles of a solar installation.

SAFETY FIRST

SOLAR PANEL CONNECTION SEQUENCE



COMPONENTS & TOOLS REQUIRED

- **COMPONENTS:**

1. **1 x Solar Panel:** (e.g., 30W to 100W, with a V_{mpp} of $\sim 18V$, which is a "12V" panel).
2. **1 x 12V Charge Controller:** (A small 10A PWM controller is perfect for this).
3. **1 x 12V Battery:** (e.g., a 12Ah to 50Ah deep-cycle battery).

4. **1 x 12V DC Load:** (e.g., a 12V 5W LED bulb and its holder).
5. **Cabling:** Red and Black solar-rated cables, plus smaller "flex" cable for the light.

- **TOOLS:**

1. Wire Stripper
2. Screwdriver
3. Digital Multimeter (DMM)

PRE-PRACTICAL SAFETY CHECKS

1. **Test the Panel:** Take the panel outside. Use the DMM (set to DC Volts) to measure its **Open Circuit Voltage (Voc)**. It should be ~20-22V.
2. **Test the Battery:** Use the DMM (set to DC Volts) to measure the **Battery Voltage**. It should be >12V (e.t., 12.6V if full).
3. **Cover the Panel:** Cover the panel with a thick blanket or cardboard. This "shuts it down" and makes it safe to wire. **Never work with a "live" panel.**

CONNECTION SEQUENCE (THE CRITICAL PROCEDURE)

This sequence is the **most important safety rule** in installation. Connecting in the wrong order can **damage the charge controller**.

STEP 1: CONNECT CHARGE CONTROLLER TO BATTERY

- **Action:** Connect the **Battery** terminals (+ and -) to the **Charge Controller's** "BATTERY" terminals (+ and -).
- **Why?** The charge controller needs to "see" the battery's voltage *first*. This is how it auto-detects if it's a 12V or 24V system. The controller's screen should light up.

STEP 2: CONNECT SOLAR PANEL TO CHARGE CONTROLLER

- **Action:** Connect the **Solar Panel's** positive (+) and negative (-) wires to the **Charge Controller's** "PV" (or "SOLAR") terminals (+ and -).
- **Why?** Now that the controller is "on" and knows it's a 12V system, it is ready to receive the ~18V from the panel.
- **Test:** Remove the blanket from the panel. The "Solar" or "Charging" light on the controller should now light up, indicating it is charging the battery.

STEP 3: CONNECT LOAD TO CHARGE CONTROLLER

- **Action:** Connect the **12V DC Light Bulb** to the **Charge Controller's** "LOAD" (or "LIGHTBULB" icon) terminals (+ and -).
- **Why?** The "Load" terminal is a special, protected output. This is the feature (from Week 8) that provides **over-discharge protection**. If the battery gets too low, the controller will automatically cut power *to this terminal* to save the battery.

STEP 4: TEST THE SYSTEM

- **Action:** Press the "load" button (if one exists) on the charge controller. The 12V light bulb should turn on, running from the battery.

DISCONNECTION SEQUENCE: To take the system apart safely, you must go in the **exact reverse order**:

1. Disconnect **PANEL** first.
2. Disconnect **LOAD** second.
3. Disconnect **BATTERY** last.

EVALUATION

1. What are the 4 essential *components* needed for this 12V DC practical?
2. What is the *very first* component you must connect to the charge controller?
3. What is the *very last* component you must connect to the charge controller?
4. Why is it critical to connect the battery *before* the solar panel?
5. What is the main safety function of the "LOAD" terminal on the charge controller?

ASSIGNMENT

1. Draw a complete, labeled wiring diagram for the practical you just performed. Show the Panel, Controller, Battery, and Load, and be sure to label all the + and - terminals correctly.
2. Write out the **Connection Sequence** (3 steps) and the **Disconnection Sequence** (3 steps).
3. What would happen if you connected the *panel* to the controller *before* the battery? (Research this - it's a common mistake).
4. What would happen if you connected the 12V light bulb *directly* to the battery terminals, bypassing the charge controller? What protection would you lose?
5. Write a 150-word "Safety Report" for this practical, listing 5 risks you identified (e.g., "Risk of short circuit," "Risk of electric shock from panel," etc.).

WEEK 11-13: REVISION & EXAMS

REVISION OF ALL SECOND TERM TOPICS

This period is for a complete review of all concepts, calculations, and practical skills from the second term.

- **Week 1 (Concepts):**
 - **Voltage (V):** Pressure. Unit: Volts.
 - **Current (I):** Flow. Unit: Amps.
 - **Resistance (R):** Friction. Unit: Ohms (Ω).
 - **Power (P):** Work. Unit: Watts (W).
 - **Ohm's Law:** $V = I \times R$
 - **Power Law:** $P = V \times I$
- **Week 2 (Tools):**
 - **Multimeter (DMM):** Measures V (Parallel), I (Series), R (Off).
 - **Clamp Meter:** Measures I (Amps) safely by clamping around one wire.
 - **Solar Tester:** Measures Irradiance (W/m^2).
 - **Hand Tools:** Strippers, Cutters, **MC4 Crimper**.
 - **PPE:** Goggles, Gloves, Boots.
- **Week 3 (Series):**
 - **Rule:** **VOLTAGE ADDS**, Current is the same.
 - **Purpose:** To increase voltage for inverters or MPPTs.
 - **Problem:** Shading ("Christmas Light Effect").
- **Week 4 (Parallel):**
 - **Rule:** **CURRENT ADDS**, Voltage is the same.
 - **Purpose:** To increase current for some controllers.
 - **Benefit:** Shade tolerant.
- **Week 6 (Mounting):**
 - **Site Assessment:** Check for **Shade**.
 - **Orientation (Nigeria):** Must face **TRUE SOUTH**.
 - **Tilt Angle:** Ideal tilt = **Latitude** of the location.

- **Types:** Roof (Flush, Tilted), Ground, Pole.
- **Week 8 (Cabling):**
 - **Sizing:** Thick cables are needed for high **Current (Amps)** or long **Distance**.
 - **Danger of thin cables:** Voltage Drop & **Fire**.
 - **Advantage of 48V:** A 48V system uses 1/4 of the current of a 12V system for the *same power*, allowing for thinner, cheaper cables.
- **Week 9 (Protection):**
 - **Earthing (Grounding):** Protects **People** from shock. Connects frames to an earth rod.
 - **Breakers/Fuses:** Protects **Wires** from **fire** (overcurrent).
 - **SPDs:** Protects **Equipment** from **lightning** (voltage spikes).
- **Week 10 (Practical):**
 - **Connection Order:** 1) Battery → Controller. 2) Panel → Controller. 3) Load → Controller.
 - **Disconnection Order:** 1) Panel. 2) Load. 3) Battery.

FINAL EXAM

The final examination will test all aspects of this term's work, including definitions, comparisons, safety rules, practical procedures, and system calculations.

SS1 SOLAR PHOTOVOLTAIC INSTALLATION & MAINTENANCE LESSON NOTES

THIRD TERM

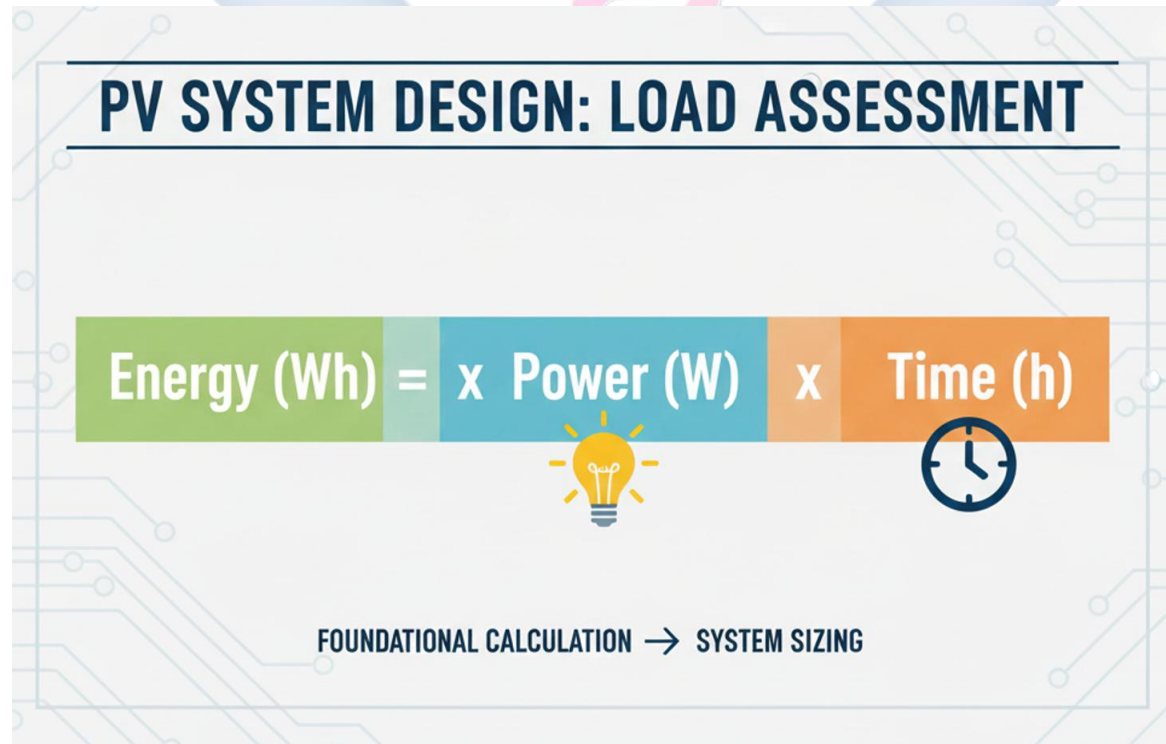
WEEK 1: PV SYSTEM DESIGN PRINCIPLES

INTRODUCTION TO SYSTEM DESIGN

Welcome to the third term. In the first and second terms, we learned what solar energy *is*, what the components *are*, and the electrical principles of how they *connect*.

This term, we learn how to be a **System Designer**. This is the most critical, "brain-work" part of the job. System design is the step-by-step process of *calculating* the perfect size and number of components (panels, batteries, inverter) to meet a specific customer's needs.

A system that is too small will fail and disappoint the customer. A system that is too large is a waste of money. This first step, **Load Assessment**, is the foundation of the entire design. If you get this wrong, the entire system will be wrong.



LOAD ASSESSMENT

A **Load** is any device or appliance that consumes electricity (e.g., a light bulb, a TV, a fan, a fridge).

A **Load Assessment** is the professional audit or survey of a customer's building to identify *every single load* they want to power, its power consumption, and how long they intend to use it.

ENERGY DEMAND CALCULATION

Our goal is to find *one single number*: the **Total Daily Energy Demand**. The unit for this is **Watt-hours (Wh) per day**.

It's crucial to understand the difference between **Power** and **Energy**:

- **POWER (W)**: This is the *rate* at which electricity is used. It is measured in **Watts (W)**. Think of this as your *speed* (e.g., 100 W).
- **ENERGY (Wh)**: This is the *total amount* of electricity used over a period of *time*. It is measured in **Watt-hours (Wh)**. Think of this as the *distance* you traveled (e.g., 100 W for 3 hours).

Formula: $Energy(Wh) = Power(W) \times Time(h)$

STEP 1: IDENTIFY ALL LOADS

Go room by room with the customer and create a list. Find the power rating (in Watts) on the sticker on the back of every appliance.

- *Example:* 10W (LED Bulb), 65W (Laptop), 80W (Fan), 100W (TV), 150W (Refrigerator).

STEP 2: DETERMINE RUN-TIME

Interview the customer to find a *realistic* average number of hours *per day* each appliance will be used.

STEP 3: CREATE THE LOAD TABLE

This is the professional way to organize the calculation.

Sample Load Assessment Table:

Appliance	Quantity (Qty)	Power (P) (Watts)	Avg. Run-Time (h) (Hours/Day)	Energy per Day (Wh) ($P \times h \times$ Qty)
LED Lights	6	10 W	5 hours	$10 \times 5 \times 6$ $= 300 \text{ Wh}$
Standing Fans	2	80 W	6 hours	$80 \times 6 \times 2$ $= 960 \text{ Wh}$
LED TV	1	100 W	4 hours	$100 \times 4 \times 1$ $= 400 \text{ Wh}$
Laptop Charger	1	65 W	2 hours	$65 \times 2 \times 1$ $= 130 \text{ Wh}$
Refrigerator	1	150 W	8 hours*	$150 \times 8 \times 1$ $= 1200 \text{ Wh}$
TOTALS	11	635 W		2,990 Wh / Day

***Important Note on Refrigerators:** A fridge is on 24/7, but its *compressor* (the motor) only runs for about 1/3 of the time (e.g., 8 hours per day). We use the *compressor run-time*, not 24 hours.

STEP 4: IDENTIFY CRITICAL, NON-CRITICAL, AND SURGE LOADS

- **Critical Loads:** Loads that *must* have power (e.g., lights, medical equipment, security system).
- **Non-Critical Loads:** Loads that are luxuries (e.g., air conditioner, electric kettle). A customer may decide they can't afford to power these.
- **Surge Loads:** Appliances with motors (fridges, pumps, ACs) that draw a huge *burst* of power (3-7x their rated power) for 1-2 seconds when they start. This is critical for sizing the inverter (Week 4).

Our final number for this customer is **2,990 Wh/Day**. This is the foundation for all calculations next week.

EVALUATION

1. What is the main difference between **Power (Watts)** and **Energy (Watt-hours)**? Use an analogy.
2. What is the primary goal of a Load Assessment? What is the final unit of measurement?
3. What is the formula for calculating Daily Energy Demand in Watt-hours?
4. A customer has 5 light bulbs (15W each) and wants to run them for 4 hours per day.
What is the *total daily energy demand* for *all* the bulbs?
5. What is a "Surge Load," and what type of appliance typically causes it?

ASSIGNMENT

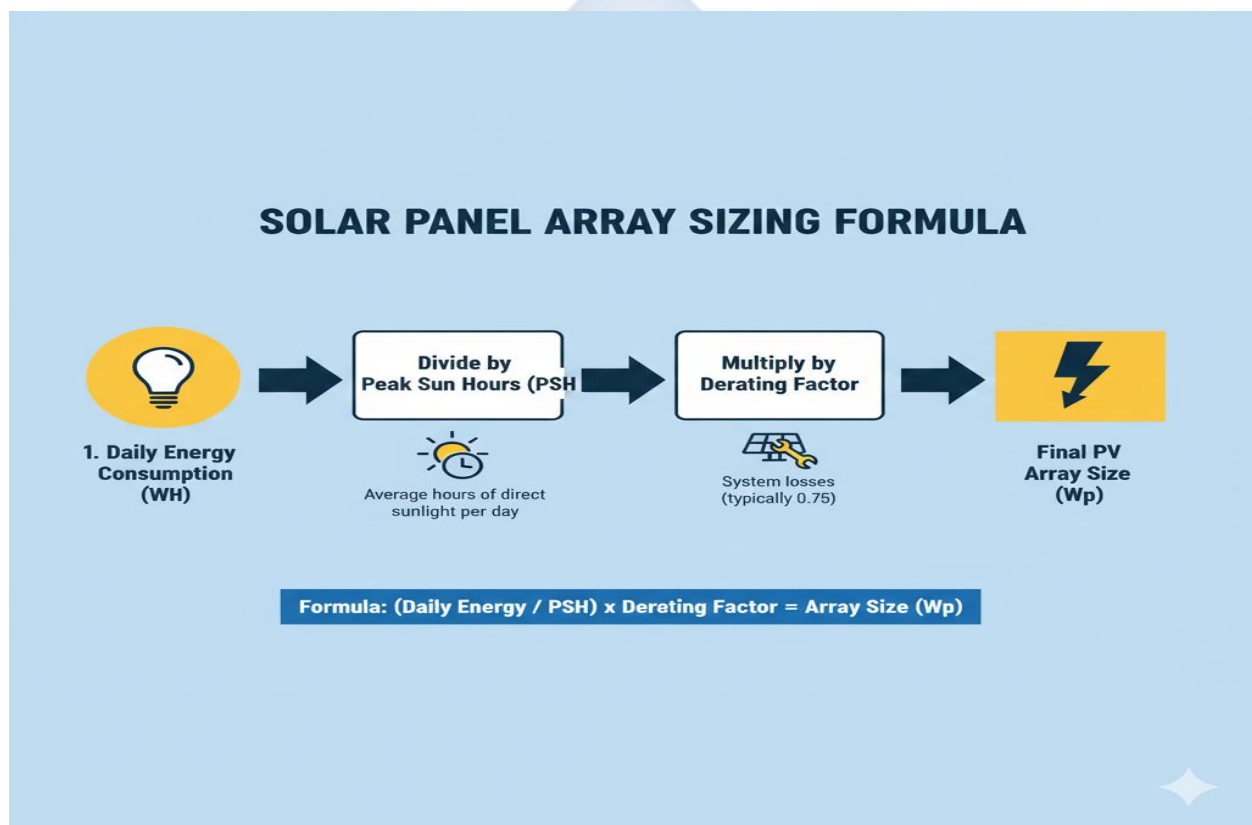
1. **Practical:** Conduct a **full load assessment** for your own home (or just your bedroom and the living room).
 - Create a Load Table (like the *sample* above).
 - List at least 6 different appliances.
 - Find their *real* power rating (in Watts) from their labels.
 - Estimate your *real* daily run-time (in hours).
 - Calculate your **Total Daily Energy Demand (Wh/Day)**.
2. Write a 200-word essay explaining the difference between a "Critical Load" and a "Non-Critical Load," giving 3 examples of each for a small business (e.g., a pharmacy).
3. A customer wants to power a 1,200W (1.5 HP) water pump for 30 minutes every day.
What is the pump's daily energy demand in Wh/Day?
4. Define the following terms: Load, Watt (W), Watt-hour (Wh), Critical Load, Compressor Cycle.
5. Why is it a bad idea to just *guess* a customer's load? What are the two risks (one for an *under-sized* system, one for an *over-sized* system)?

WEEK 2: SIZING SOLAR PANELS (THE SOLAR ARRAY)

INTRODUCTION

Last week, we determined our customer's **Total Daily Energy Demand (2,990 Wh/Day)**. This is the "target." This week, we size the "engine" that will *generate* that energy: the **Solar Array** (the total group of solar panels).

Our goal is to calculate the **Total Watts-peak (W_p)** of panels needed to reliably generate 2,990 Wh in *one* average day.



STEP 1: UNDERSTANDING THE "FUEL" - PEAK SUN HOURS (PSH)

- **Definition:** We cannot just use "hours of daylight." The sun is weak in the morning, strong at noon, and weak in the evening.
- **Peak Sun Hours (PSH)** is a standard value that represents the *equivalent* number of hours per day that the sun was shining at its *peak intensity* (1000 W/m^2).

- It's the "average daily solar fuel" for a location. This number is the *most important* variable in solar design. It is found from your **Site Assessment** (from 2nd Term).
- **Example (Nigeria):**
 - Lagos (Coastal): $\approx 4.5 - 5.0$ PSH (more clouds/haze)
 - Kano (Northern): $\approx 6.0 - 6.5$ PSH (more intense sun)
- **For our calculation, let's assume our customer is in Lagos and has 5.0 PSH.**

STEP 2: CALCULATE THE "RAW" ARRAY SIZE

This is the basic calculation.

- **Formula:** Raw Array Size (W_p) = Total Daily Energy (Wh/Day) / Peak Sun Hours (h)
- **Calculation:**
 - $2,990 \text{ Wh} / 5.0 \text{ h} = \mathbf{598 W_p}$

This means, in a *perfect world*, a 598W solar array would generate exactly 2,990 Wh of energy in that location.

STEP 3: ACCOUNT FOR REAL-WORLD LOSSES (THE "DERATING FACTOR")

The "perfect world" does not exist. A solar panel *never* produces 100% of its sticker rating. We must "oversize" the array to account for all the real-world inefficiencies. This is called the **Derating Factor**.

- **Where is energy lost?**
 - a. **Heat:** Solar panels are rated at 25°C. A hot roof in Nigeria can be 60-70°C. High heat *reduces* a panel's power output (a "temperature coefficient loss").
 - b. **Dirt/Dust:** A layer of dust or bird droppings on the panel blocks sunlight.
 - c. **Wire Losses:** Energy is lost as heat (Resistance) in the DC cables from the roof to the inverter.
 - d. **Component Mismatch:** Not all panels are 100% identical.
- **The Factor:** To account for all these losses, we apply a safety factor. A standard Derating Factor is **1.3** (meaning we oversize the array by 30%).

- **Formula: Final Array Size (W_p) = Raw Array Size (W_p) $\times$ Derating Factor**
- **Calculation:**
 - $598 \text{ W} \times 1.3 = 777.4 \text{ W}_p$

STEP 4: CHOOSE THE ACTUAL PANELS

- We must now buy *at least* **777.4 Watts** of solar panels.
- We look at what is available in the market (e.g., 350W, 400W, 450W panels).
- **Option 1:** Using 350W panels.
 - $777.4 \text{ W} / 350 \text{ W} = 2.22$ panels. We must **always round up**.
 - **Choice: Three 350W panels.** (Total = 1050 W_p).
- **Option 2:** Using 400W panels.
 - $777.4 \text{ W} / 400 \text{ W} = 1.94$ panels. We must **always round up**.
 - **Choice: Two 400W panels.** (Total = 800 W_p).

Conclusion: The **two 400W panels** (800 W_p total) is the most efficient choice that meets our minimum requirement of 777.4 W_p . This is our final solar array size.

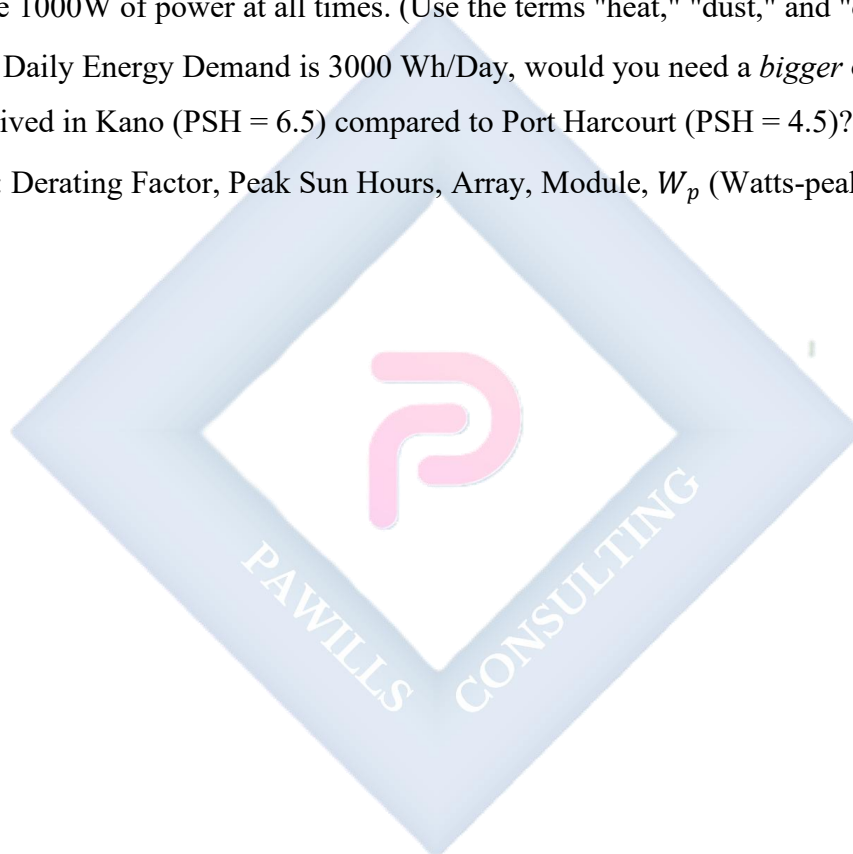
EVALUATION

1. What is "Peak Sun Hours" (PSH), and why is it different from "hours of daylight"?
2. What is the "Derating Factor," and why must we use it? (List 3 real-world losses it accounts for).
3. What is the main formula for calculating the *Raw* Array Size (before derating)?
4. A customer needs 4,000 Wh/Day. Their location has 5.0 PSH. Using a 1.3 derating factor, what is the *Final* Array Size (in W_p) they need?
5. Based on your answer to Q4, how many 300W panels would you need to install?

ASSIGNMENT

1. **Calculate Your Own Solar Array:**

- (a) Take your **Total Daily Energy Demand (Wh/Day)** from your Week 1 Assignment.
 - (b) Assume your location has **5.0 PSH**.
 - (c) Assume a **Derating Factor of 1.3**.
 - (d) Calculate the **Final Array Size (W_p)** you need.
2. Based on your calculation in Q1, how many **450W panels** would you need to buy?
 3. Write a 200-word essay explaining to a customer why a 1000W array of panels will *never* produce 1000W of power at all times. (Use the terms "heat," "dust," and "derating").
 4. If your Daily Energy Demand is 3000 Wh/Day, would you need a *bigger* or *smaller* array if you lived in Kano (PSH = 6.5) compared to Port Harcourt (PSH = 4.5)? Explain why.
 5. Define: Derating Factor, Peak Sun Hours, Array, Module, W_p (Watts-peak).

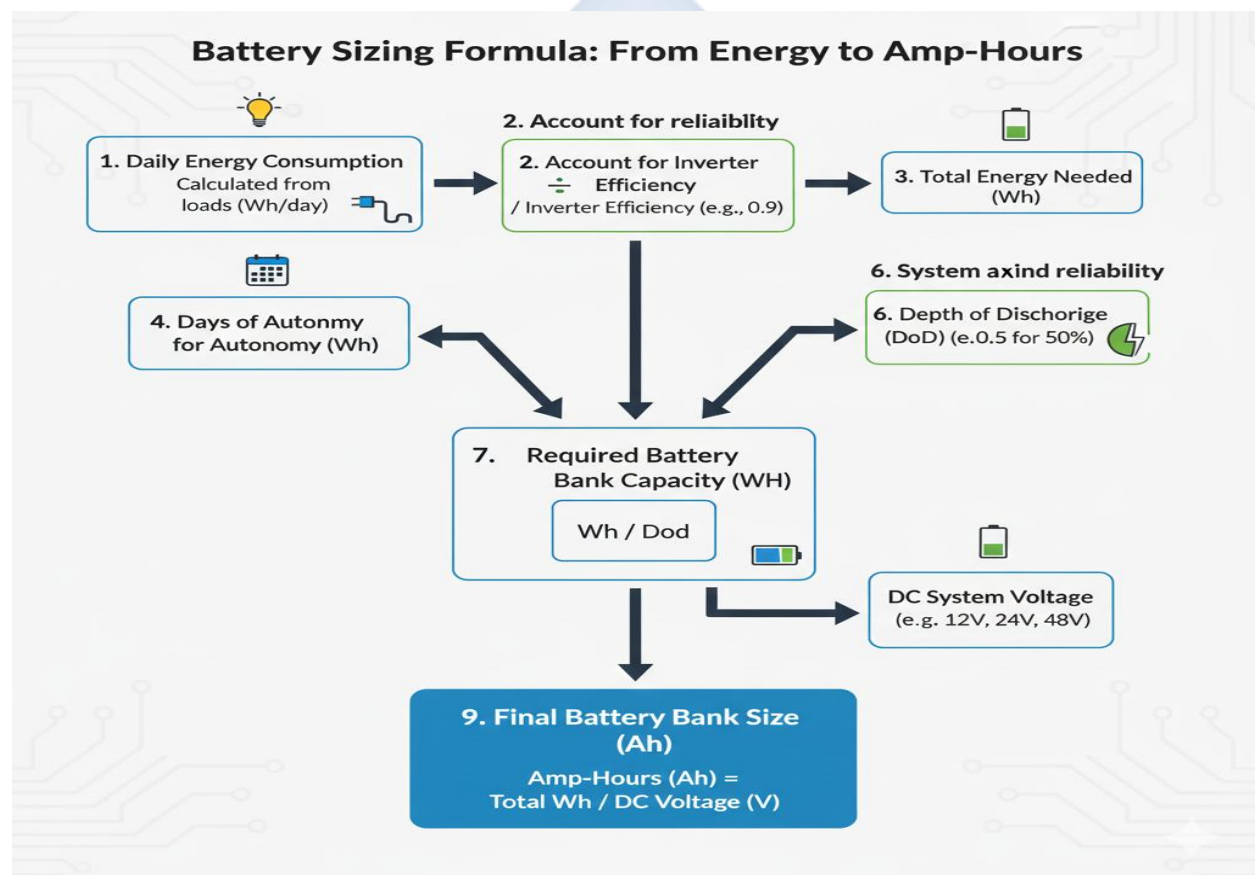


WEEK 3: SIZING BATTERIES

INTRODUCTION

Last week, we sized our "engine" (the **800 W_p Solar Array**). This week, we size our "storage tank": the **Battery Bank**.

The battery bank's job is to store the energy from the panels during the day to power our loads (2,990 Wh/Day) at night, and also to provide power during cloudy or rainy days. This calculation is the *most complex* and *most expensive* part of the design.



STEP 1: ACCOUNT FOR SYSTEM INEFFICIENCIES

- The components are not 100% efficient. The inverter (which we'll size next week) wastes power (as heat) when it converts DC (from the battery) to AC (for the appliances).
- We must "oversize" our battery storage to account for this loss. A typical **Inverter Efficiency** is 85% - 90%. We will use **90% (or 0.90)**.

- **Formula:** Adjusted Daily Load (Wh) = Total Daily Energy (Wh/Day) / Inverter Efficiency
- **Calculation:**
 - $2,990 \text{ Wh} / 0.90 = \mathbf{3,322 \text{ Wh/Day}}$
- This is our new "target" number. We need to be able to *store and retrieve* 3,322 Wh for every *one* day of power.

STEP 2: DETERMINE "DAYS OF AUTONOMY"

- **Definition:** This is the number of days you want the system to run with **ZERO SUN**. It's the "rainy day" safety buffer.
- **Standard Values:**
 - **1 Day:** For a budget system (common).
 - **2 Days:** A very reliable, standard system.
 - **3 Days:** For a critical-load system (e.g., a hospital, a remote site).
- **Formula:** Total Usable Storage (Wh) = Adjusted Daily Load (Wh) × Days of Autonomy
- **Calculation (Using 1 Day of Autonomy):**
 - $3,322 \text{ Wh/Day} \times 1 \text{ Day} = \mathbf{3,322 \text{ Wh}}$
- This is the *usable energy* we must be able to pull from our batteries.

STEP 3: ACCOUNT FOR DEPTH OF DISCHARGE (DOD)

This is the *most important concept* in battery health and sizing.

- **Definition:** You cannot (and must not) drain a battery to 0%. **Depth of Discharge (DoD)** is the *maximum percentage* of the battery's total capacity you can *safely* use without permanently damaging it.
- **Standard Values:**
 - **Flooded Lead-Acid Batteries (FLA): 50% DoD** (You can only use *half* the battery).
 - **Lithium-ion (LiFePO4) Batteries: 80% or 90% DoD** (You can use most of it).

- This is why Lithium is more expensive—a 100Ah Lithium battery has almost the same *usable* power as a 200Ah Lead-Acid battery.
- **Formula:** Total Required Capacity (Wh) = Total Usable Storage (Wh) / DoD
- **Calculation (Assuming we use Lead-Acid batteries at 50% DoD):**
 - $3,322 \text{ Wh (Usable)} / 0.50 \text{ (DoD)} = \mathbf{6,644 \text{ Wh}}$
- This is the *total rated capacity* of the battery bank we must buy.

STEP 4: DETERMINE SYSTEM VOLTAGE (12V, 24V, OR 48V)

- Before we buy, we must choose a **System Voltage**. This is the DC voltage of our battery bank and inverter.
- **Rule of Thumb:**
 - Small systems (<1000W) → 12V
 - Medium systems (1000W - 3000W) → 24V
 - Large systems (>3000W) → 48V
- Since our loads are significant, a 12V system would have very high current. **We will choose a 24V system.**

STEP 5: CONVERT WATT-HOURS (WH) TO AMP-HOURS (AH)

- Batteries are *sold* in **Volts (V)** and **Amp-Hours (Ah)** (e.g., "12V, 200Ah"). We must convert our Watt-hour requirement into Amp-hours to know what to buy.
- **Formula (from Power Law):** Capacity (Ah) = Total Required Capacity (Wh) / System Voltage (V)
- **Calculation:**
 - $6,644 \text{ Wh} / 24\text{V} = \mathbf{276.8 \text{ Ah}}$

STEP 6: CHOOSE THE BATTERIES

- **Our Final Size:** We need a **24V, 276.8 Ah** (or larger) Lead-Acid battery bank.
- **How to build it?** A standard solar battery is the "12V, 200Ah" battery.
 - a. To get 24V: We must connect **two** 12V batteries in **Series**.

- b. This gives us: $(12V + 12V) = 24V$. The current stays the same: **200Ah**.
 - c. This **24V, 200Ah** bank is *too small* (we need 276.8Ah).
 - d. So, we must *add another string in parallel*.
 - e. We take **four 12V, 200Ah** batteries total.
 - f. We build two strings of 2 in series (String 1 = 24V, 200Ah).
 - g. We build another string of 2 in series (String 2 = 24V, 200Ah).
 - h. We connect those two strings in **parallel**.
 - i. **Final Bank:** Voltage stays the same (24V). Current adds ($200Ah + 200Ah = 400Ah$).
- **Conclusion:** The **24V, 400Ah** bank (made of four 12V/200Ah batteries) is larger than our 276.8Ah requirement. This is the correct, safe choice.

EVALUATION

1. What is "Days of Autonomy," and why is it an important safety buffer?
2. Define "Depth of Discharge (DoD)." What is the standard DoD for (a) Lead-Acid and (b) Lithium-ion batteries?
3. A customer needs 2,000 Wh of *usable* storage. If they use Lithium batteries (80% DoD), what is the *total rated capacity* (in Wh) they must buy?
4. What is the formula to convert Watt-hours (Wh) to Amp-hours (Ah)?
5. A 24V, 400Ah battery bank is made of 12V, 200Ah batteries. How many batteries are needed, and how are they connected? (Draw a simple diagram).

ASSIGNMENT

1. **Calculate Your Own Battery Bank:**
 - (a) Take your **Total Daily Energy Demand (Wh/Day)** from your Week 1 Assignment.
 - (b) Assume 90% (0.90) Inverter Efficiency.
 - (c) Assume **1 Day of Autonomy**.
 - (d) Assume you will use **Lead-Acid** batteries (50% DoD).

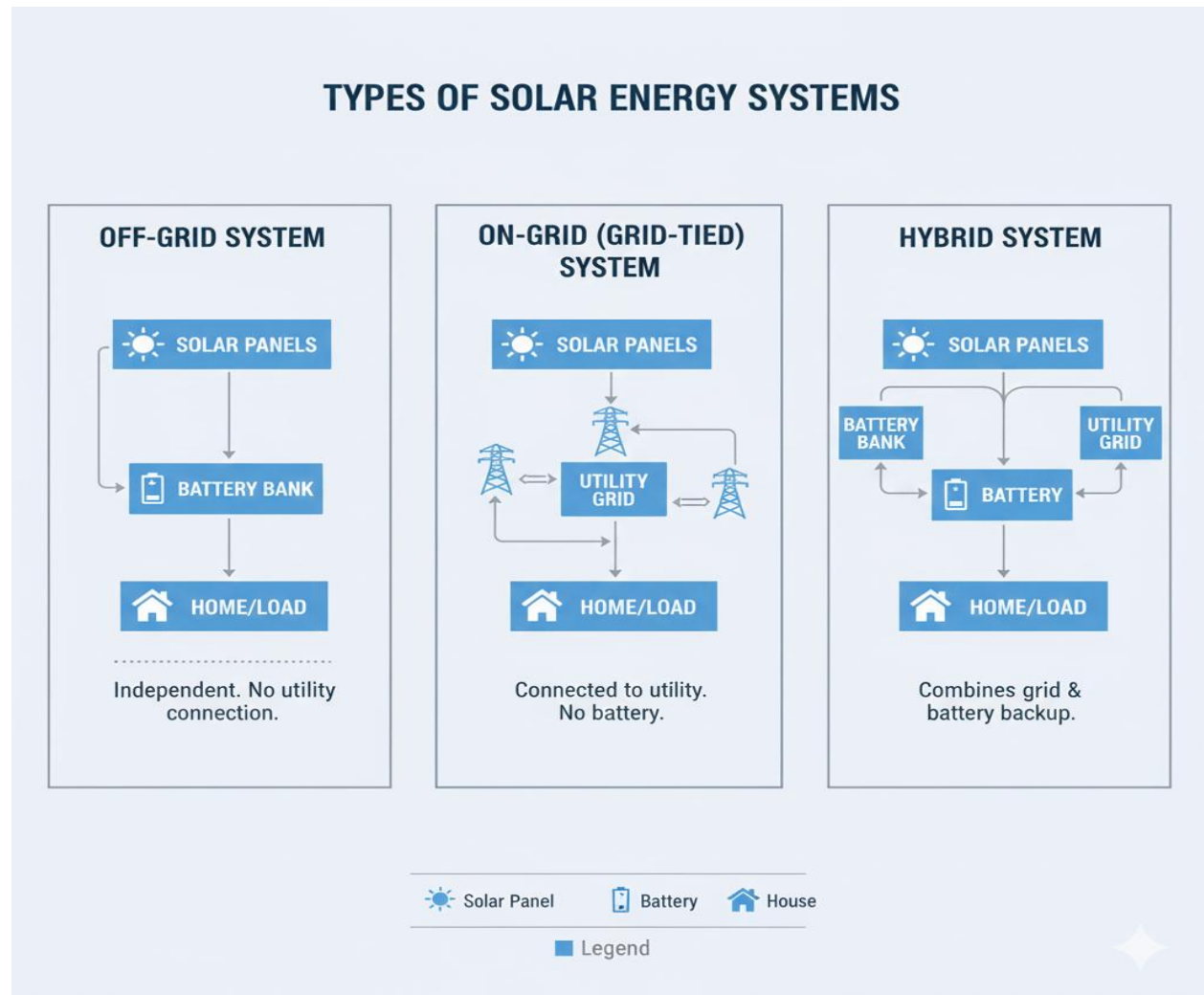
- (e) Assume a **12V** System Voltage.
 - (f) Calculate your **Final Required Amp-Hours (Ah)**.
2. Based on your calculation in Q1, how many **12V, 200Ah** batteries would you need to buy?
 3. Re-do the calculation from Q1, but this time assume you are using **Lithium-ion** batteries (90% DoD). How many Amp-Hours (Ah) do you need now?
 4. Draw a wiring diagram for the final battery bank in the lesson's example (four 12V, 200Ah batteries connected in a series-parallel arrangement to make a 24V, 400Ah bank).
 5. Define: Days of Autonomy, DoD, System Voltage, Amp-Hour, Inverter Efficiency.



WEEK 4: SIZING CHARGE CONTROLLERS & INVERTERS

INTRODUCTION

We have now sized our "loads" (2,990 Wh/Day), our "engine" (800 W_p Array), and our "storage tank" (24V, 400Ah Bank). The final step is to size the "brains" and the "heart" of the system: the **Inverter** and the **Charge Controller**.



SIZING THE INVERTER

The **Inverter** (which converts battery DC to household AC) is sized based on two factors: the *loads it must run* and the *battery it connects to*.

1. SIZING FOR VOLTAGE (THE BATTERY MATCH)

- This is the simplest, non-negotiable rule. The inverter's **DC Input Voltage** *must* match the **System Voltage** of the battery bank.
- **Our Example:** We chose a **24V** battery bank.
- **Conclusion:** We **must** buy a **24V Inverter**.

2. SIZING FOR WAVEFORM (PURITY)

- **Modified Sine Wave:** A "choppy" wave. It is cheap but can damage sensitive electronics (laptops, TVs) and makes motors (fans, fridges) run hot.
- **Pure Sine Wave (PSW):** A clean, smooth wave, identical to grid power. It is safe for all appliances.
- **Conclusion:** For any modern household, a **Pure Sine Wave** inverter is the only professional choice.

3. SIZING FOR POWER (WATTS / KVA)

This is the main "size" (e.g., 1.5kW, 3.5kVA, 5kW). It must be large enough to handle all the loads.

- **Step A: Continuous Load:** The sum of all loads that might run *at the same time*.
 - **From our Load Table (W1):** (6x Lights) + (2x Fans) + (1x TV) + (1x Laptop) + (1x Fridge)
 - $(60W) + (160W) + (100W) + (65W) + (150W) = \mathbf{535 \text{ Watts}}$
 - We must add a **safety factor** (e.g., 25%) so the inverter isn't running at 100% capacity.
 - $535 \text{ W} \times 1.25 = \mathbf{669 \text{ W}}$ (This is our minimum *continuous* rating).
- **Step B: Surge Load (The "Kick"):**
 - The inverter must be able to handle the *startup surge* of all motors.
 - **Our Load:** The 150W Refrigerator. A fridge has a surge of $\approx 6\text{-}7\times$ its running power.
 - **Fridge Surge:** $150 \text{ W} \times 7 = \mathbf{1050 \text{ W}}$.
 - **Total Surge:** 1050 W (fridge) + (all other loads, 385 W) = $\approx \mathbf{1435 \text{ W}}$.

- **Conclusion:** We need an inverter that can handle a *continuous* load of at least 669 W and a *surge* of at least 1435 W.
- **Final Choice:** Inverters are sold in standard sizes. A **1500W (or 1.5kW)** inverter is a safe and standard choice. It meets our 669W continuous need and our 1435W surge need.
- **FINAL INVERTER SIZING: 1.5kW, 24V, Pure Sine Wave Inverter.**

SIZING THE CHARGE CONTROLLER

The **Charge Controller** (which protects the battery from the panels) is sized based on the **Solar Array** and the **Battery Bank**.

1. CHOOSING THE TYPE (PWM VS. MPPT)

- **Rule:** If the Panel Array's nominal voltage is *different* from the Battery Bank's voltage, you **must use an MPPT**.
- **Our Example:** Our array (W2) consists of **two 400W panels**. A 400W panel is a high-voltage panel (e.g., $V_{mpp} \approx 40V$). Our battery bank is **24V**.
- **Conclusion:** Since $40V \neq 24V$, we **must use an MPPT controller**.

2. SIZING FOR VOLTAGE (V_{OC})

- The controller's **Max PV Input Voltage** *must* be higher than the array's **Open Circuit Voltage** (V_{OC}), especially in series.
- **Our Example:** Our 400W panel has a V_{OC} of $\approx 49V$.
- If we connect our two panels in **Series**: $V_{OC} = 49V + 49V = 98V$.
- If we connect them in **Parallel**: $V_{OC} = 49V$.
- **Conclusion:** We must choose a controller with a V_{OC} rating *higher* than our string. A **100V** or **150V** controller is a standard, safe choice.

3. SIZING FOR CURRENT (AMPS)

- This is the main rating (e.g., 20A, 40A, 60A). The controller must be able to handle the maximum current it will *output* to the battery.
- **Formula (for MPPT):** **Controller Amps (A) = Total Array Power (W) / Battery Voltage (V)**

- **Calculation:**
 - Our Array Power (W_2) = 800 W_p
 - Our Battery Voltage (W_3) = 24V
 - $I = P/V \rightarrow 800 \text{ W}/24 \text{ V} = \mathbf{33.3}$ Amps
- This (33.3A) is the *maximum charging current* the MPPT will send to the battery.
- We must add a **safety factor** (e.g., 25%).
- $33.3 \text{ A} \times 1.25 = \mathbf{41.6}$ Amps
- **Conclusion:** We must buy a controller that is *at least* 41.6A.
- **FINAL CONTROLLER SIZING:** A **50A, 100V MPPT Charge Controller** is the correct professional choice.

EVALUATION

1. What are the three main factors you must consider when sizing an inverter? (Voltage, Waveform, Power).
2. A fridge has a 100W running power and a 600W surge. A TV is 80W. What is the *minimum continuous* rating and *minimum surge* rating for an inverter that can power them both?
3. An inverter is labeled "48V." What does this mean?
4. What is the main formula for calculating the required **Amperage** of an *MPPT* charge controller?
5. Why *must* we use an MPPT controller (and not a PWM) if our panel array has a V_{mpp} of 72V and our battery bank is 24V?

ASSIGNMENT

1. **Calculate Your Own Inverter & Controller:**
 - (a) Take your **Load Assessment (W1)** and **Battery Bank Voltage (W3)**.
 - (b) Calculate the **Continuous Wattage** of all your loads.
 - (c) Assume one load is a motor with a 6x surge. Calculate the **Surge Load**.
 - (d) Choose a suitable **Inverter (W and V)**.

- (e) Use your **Array Size (W2)** and **Battery Voltage (W3)** to calculate the required **Charge Controller Amperage**.
 - (f) Choose a suitable **Charge Controller (A and V)**.
2. Write a 200-word essay explaining to a new client why their new 1000W inverter *cannot* power their 800W water pump. (Hint: Use the word "surge").
 3. Research a popular inverter brand (e.g., Victron, Outback, Su-Kam, Luminous). Find the "spec sheet" for a 3kW (3000W) inverter. What is its DC input voltage? What is its surge rating?
 4. Why is it a very bad idea to oversize an inverter *too much*? (e.g., using a 5000W inverter to power a 30W load). (Hint: Research "inverter standby power" or "self-consumption").
 5. Define: Continuous Load, Surge Load, V_{OC} (Open Circuit Voltage), MPPT, Pure Sine Wave.



CASE STUDY: THE OKORO FAMILY'S BUNGALOW

CLIENT: The Okoro Family. **LOCATION:** A rural home near Benin City, Edo State.

PROBLEM: No grid power (NEPA). They rely on a noisy, expensive generator. **GOAL:** An off-grid system for essential loads. **SITE ASSESSMENT DATA:**

- **Shade:** The roof is 100% shade-free.
- **Orientation:** The roof faces **True South**.
- **Peak Sun Hours (PSH):** ≈ 4.5 PSH (due to coastal/forest weather).

STEP 1: LOAD ASSESSMENT

The family has agreed on the following *critical loads*:

Appliance	Qty	Power (W)	Hours/Day (h)	Energy (Wh/Day)
LED Lights	8	9 W	5 hours	$9 \times 5 \times 8$ $= 360 \text{ Wh}$
Standing Fans	3	70 W	6 hours	$70 \times 6 \times 3$ $= 1260 \text{ Wh}$
LED TV + DSTV	1	120 W	4 hours	$120 \times 4 \times 1$ $= 480 \text{ Wh}$
Small Fridge	1	100 W	8 hours	$100 \times 8 \times 1$ $= 800 \text{ Wh}$
TOTALS		502 W		2,900 Wh / Day

- **Total Daily Energy Demand** = 2,900 Wh/Day.
- **Total Continuous Load** = 502 W.
- **Surge Load:** The 100W Fridge $\times 7 = 700 \text{ W}$.

STEP 2: SIZING THE BATTERY BANK

1. **System Voltage:** The load is moderate. A **24V** system is a good choice.
2. **Adjust for Inverter Loss (90% Eff):**
 - $2,900 \text{ Wh} / 0.90 = 3,222 \text{ Wh/Day}$

3. **Add Days of Autonomy (2 Days):** They have no grid, so a 2-day buffer is wise.
 - $3,222 \text{ Wh} \times 2 \text{ Days} = \mathbf{6,444 \text{ Wh (Usable)}}$
4. **Adjust for DoD (50% for Lead-Acid):**
 - $6,444 \text{ Wh} / 0.50 = \mathbf{12,888 \text{ Wh (Total Rated)}}$
5. **Convert to Amp-Hours (Ah):**
 - $12,888 \text{ Wh} / 24\text{V (System Voltage)} = \mathbf{537 \text{ Ah}}$
6. **Choose Batteries (12V, 200Ah):**
 - We need a **24V, 537Ah** bank.
 - *Series:* $2 \times 12\text{V batteries} = 24\text{V}, 200\text{Ah}$ (Too small)
 - *Series-Parallel:* We need to parallel 3 strings. $(200\text{Ah} + 200\text{Ah} + 200\text{Ah} = 600\text{Ah})$.
 - **Final Choice: Six 12V, 200Ah batteries.**
 - (3 strings in parallel, with 2 batteries in series per string).
 - **Final Bank: 24V, 600Ah.**

STEP 3: SIZING THE SOLAR ARRAY

1. **Energy Target:** 3,222 Wh/Day (from Step 2.1).
2. **Peak Sun Hours (PSH):** 4.5 PSH (from Site Assessment).
3. **Raw Array Size:**
 - $3,222 \text{ Wh} / 4.5 \text{ h} = \mathbf{716 \text{ W}_p}$
4. **Adjust for Derating Factor (1.3):**
 - $716 \text{ W} \times 1.3 = \mathbf{930.8 \text{ W}_p}$
5. **Choose Panels (e.g., 330W Panels):**
 - $930.8 \text{ W} / 330 \text{ W} = 2.8 \text{ panels}$. → We must use **3 panels**.
 - **Final Array: Three 330W panels** (Total = 990 W_p).

STEP 4: SIZING THE INVERTER

1. **Voltage:** Must match the battery bank. **24V**.

2. **Waveform:** Must be **Pure Sine Wave** (for the TV and Fridge).
3. **Continuous Power:**
 - Total Loads = 502 W.
 - $502 \text{ W} \times 1.25$ (Safety Factor) = 627.5 W.
4. **Surge Power:**
 - Fridge Surge (700 W) + Other Loads (402 W) = 1102 W.
5. **Final Choice:** A 1.5kW (1500W) inverter is a standard, safe choice.
 - **Final Inverter: 1.5kW, 24V, Pure Sine Wave Inverter.**

STEP 5: SIZING THE CHARGE CONTROLLER

1. **Type:** Panel voltage (a 330W panel is $\approx 38\text{V } V_{mpp}$) is higher than the battery (24V). Must be **MPPT**.
2. **Voltage (V_{OC}):** A 330W panel has a $V_{OC} \approx 46\text{V}$.
 - If we connect all 3 panels in *series*: $V_{OC} = 46 + 46 + 46 = 138\text{V}$.
 - A **150V** controller is required.
3. **Current (Amps):**
 - $I = \text{Total Array Power} / \text{Battery Voltage}$
 - $I = 990 \text{ W} / 24 \text{ V} = \mathbf{41.25 \text{ Amps}}$
4. **Safety Factor:** $41.25 \text{ A} \times 1.25 = 51.5 \text{ A}$.
5. **Final Choice:** A **60A, 150V MPPT Charge Controller** is the correct choice.

EVALUATION

1. What was the *first* calculation in this entire case study, and why?
2. Why did we choose 2 Days of Autonomy for this battery calculation?
3. How many 12V, 200Ah batteries were needed, and how were they wired?
4. Why did we *have* to choose an MPPT controller, not a PWM?
5. What was the *determining factor* for the Inverter's size: the continuous load or the surge load?

ASSIGNMENT

1. **Your Turn:** Design a system for a **Small Barbing Salon**.
 - **Location:** Abuja (PSH = 5.5).
 - **Loads:**
 - 2 x Clippers (40W total) for 6 hours/day
 - 2 x Fans (80W each) for 6 hours/day
 - 1 x TV (100W) for 6 hours/day
 - 4 x Lights (10W each) for 6 hours/day
 - **Design Goals:** 12V System, 1 Day Autonomy, Lead-Acid (50% DoD).
2. **For the Salon:** Calculate the **Total Daily Energy Demand (Wh/Day)**.
3. **For the Salon:** Calculate the **Final Battery Bank Size (in Ah)**. (Use 90% Inverter Eff).
4. **For the Salon:** Calculate the **Final Solar Array Size (in W_p)**. (Use 1.3 Derating Factor).
5. **For the Salon:** Size the **Inverter (W, V, Type)** and the **Charge Controller (A, V, Type)**. (Assume a 10W Clipper has a 40W surge; 330W panels (V_{OC} 46V)).

WEEK 8: PV SYSTEM INSTALLATION STEPS I

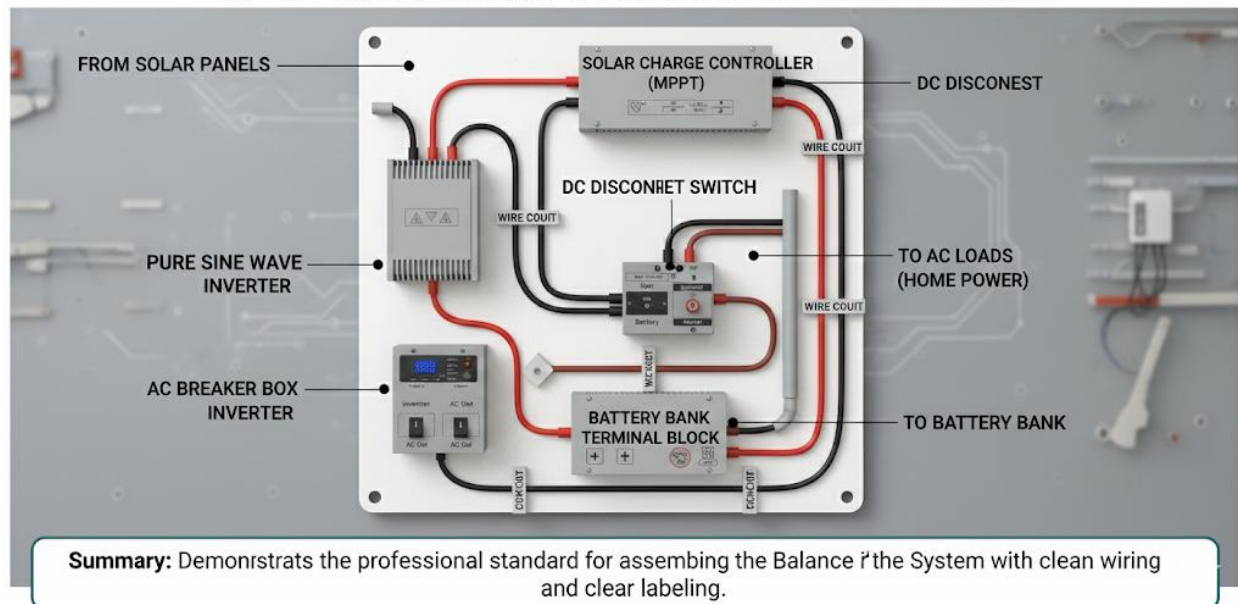
TOPIC: MOUNTING PANELS & INSTALLING BALANCE OF SYSTEM

INTRODUCTION

The "design" phase is complete. We now move to the "installation" phase. This is the physical, "hands-on" work of building the system we designed. This week covers the *mechanical* installation: attaching the hardware.

Safety is the #1 priority. All work on a roof or with heavy equipment requires Personal Protective Equipment (PPE): **hard hats, safety glasses, gloves, and safety boots.**

SOLAR INSTALLATION I: COMPONENT BOARD LAYOUT



STEP 1: SITE PREPARATION & SAFETY

1. **Clear the Area:** Remove all debris from the roof work area and the indoor component board area.
2. **Roof Safety:** For pitched (angled) roofs, secure ladders and use safety harnesses if the roof is high or steep.
3. **Confirm the Plan:** Have the design document (from Week 6) and the Site Assessment (from Week 4) in hand.

4. **Marking:** Use chalk lines to mark *exactly* where the mounting rails will go, ensuring they are aligned with the roof's structural **rafters** or "trusses."

STEP 2: MOUNTING THE SOLAR PANELS (THE ARRAY)

The mounting method depends entirely on the **roof type**.

A. *PITCHED (ANGLED) METAL "ZINC" ROOFS*

1. **Locate Rafters:** Tap the roof to find the solid wood rafters (trusses) *under* the metal sheet. You *must* anchor into these.
2. **Install L-Feet:** Bolt **L-Feet** (angle brackets) *through* the metal roof and *into* the rafters.
3. **Waterproofing:** A rubber **flashing** or high-grade sealant (like Butyl) *must* be applied around the bolt hole *under* the L-Foot to prevent leaks.
4. **Install Rails:** Attach the long aluminum **solar rails** to the L-Feet.
5. **Install Panels:** Lay the solar panels onto the rails and secure them using **mid-clamps** (between panels) and **end-clamps** (at the ends of the array).
6. **Grounding:** Run a **bare copper wire (grounding wire)** connecting the *metal frame* of *every single panel* and *every rail* to each other, using "grounding lugs."

B. *FLAT (CONCRETE) ROOFS*

1. **Assemble Racks:** Assemble the **A-Frame (Tilted) Mounts** on the ground.
2. **Orientation:** Place the frames on the roof, using a **compass** to ensure they face **TRUE SOUTH**.
3. **Tilt Angle:** Use a **clinometer (angle finder)** to set the frame's tilt angle to the correct **latitude** (e.g., 10°).
4. **Ballasting (Securing):** *Do not* drill holes in a flat roof. Instead, use **ballast**. Place heavy, cast-concrete blocks onto the base of the frames. This weight (ballast) holds the array down against the wind.
5. **Install Panels:** Clamp the panels to these frames.

STEP 3: INSTALLING THE BALANCE OF SYSTEM (BOS)

BOS refers to all the *other* components of the system that are not the panels. This is the "component board" or "power wall" installed indoors.

1. **Location:** Choose a location that is **cool, dry, well-ventilated, and easily accessible**. *Never* in a bedroom, dusty room, or unventilated closet.
2. **Mounting Board:** Attach a large, non-flammable board (like "marine plywood") to the wall. This makes the installation neat and organized.
3. **Mount Components:** Securely mount the main components to this board, leaving space between them for air circulation and heat dissipation.
 - **Inverter** (heavy, must be well-anchored)
 - **Charge Controller**
 - **Breaker Box / Combiner Box**
4. **Battery Storage:** Place the batteries in their final position. This must be **on a low, strong rack or insulated stand**. *Never* place batteries directly on a concrete floor, as this will drain their charge and shorten their life.

EVALUATION

1. What does "PPE" stand for, and what 4 items must an installer wear?
2. When mounting on a pitched "zinc" roof, what must you anchor the L-Foot into?
3. What is "Ballast," and on what type of roof is it used?
4. What is "BOS" (Balance of System)?
5. Why must you *never* install batteries directly on a concrete floor?

ASSIGNMENT

1. Draw a simple, labeled diagram of a "Pitched Roof" mounting system, showing the Rafter, L-Foot, Rail, and Panel.
2. Draw a simple, labeled diagram of a "Flat Roof" mounting system, showing the A-Frame (Tilted Mount) and the Ballast Blocks.

3. Write a 200-word safety report on the "Top 5 Dangers of Working on a Roof" and the safety precautions for each.
4. What is a "grounding lug," and what is its purpose?
5. Research "Mid-Clamps" and "End-Clamps." What is the difference, and where is each one used?



WEEK 9: PV SYSTEM INSTALLATION STEPS II

TOPIC: ELECTRICAL WIRING & CONNECTION

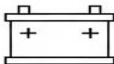
INTRODUCTION

This is the most dangerous and most critical step of the installation. The *sequence* of connection is not optional—it is a *safety rule* designed to protect the equipment (especially the charge controller) and the installer.

SAFETY FIRST: All breakers must be in the **OFF** position. All panel strings must be "open" (unconnected) or covered. All wiring must follow **Polarity** (Red for $\oplus$, Black for $\ominus$).

Installation II: (Wiring & Connection)

 **THE GOLDEN RULE OF CONNECTION**

- 1. CONNECT BATTERY FIRST**

- 2. CONNECT PV (SOLAR PANELS)**

- 3. CONNECT LOADS (APPLIANCES)**


WRONG ORDER CAN DAMAGE EQUIPMENT!

Always follow this sequence to protect your charge controller and devices.

STEP 1: RUN THE WIRING

1. Earthing (Grounding):

- This is the *first* wire to run.

- Run a bare copper or green/yellow **Ground Wire** from the rooftop array (panels and rails) to the **Earth Rod** (a 6-8ft copper rod driven deep into the ground).
- Run another ground wire from the **Inverter chassis (metal case)** and the **component board ground busbar** to the *same* Earth Rod.

2. DC Wiring (PV Array):

- Connect the panels on the roof into their correct **Series/Parallel strings** (as designed in Week 3).
- Run the main $\oplus$ and $\ominus$ **solar cables** (UV-resistant) from the roof to the **PV Breaker** (DC Disconnect) on the component board.

3. Battery Bank Wiring:

- Build the battery bank (e.g., in series-parallel) as designed (Week 8).
- Run the main *thick* **battery cables** from the bank's main terminals to the **Main Battery Breaker/Fuse**.

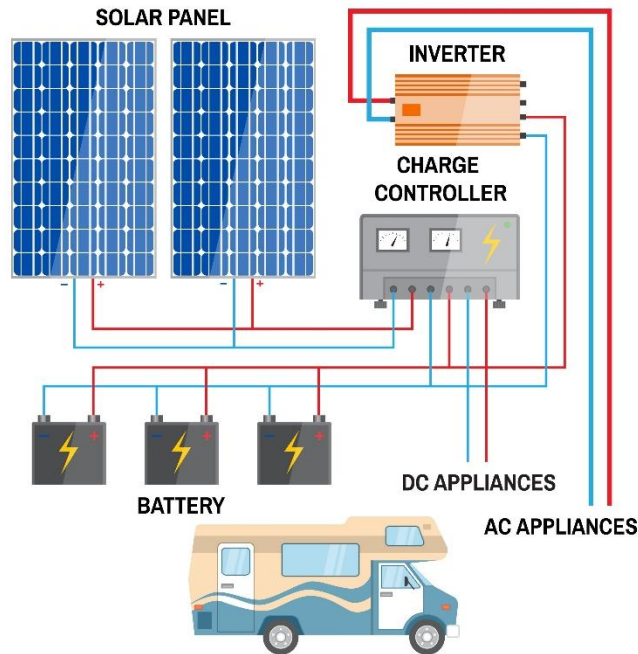
4. AC Wiring:

- Run the **AC cable** (Live, Neutral, Earth) from the **Inverter "AC-OUT"** terminal to the **AC Breaker** in the new distribution board.
- Connect the house's "critical load" circuits (lights, sockets) to this AC breaker.

STEP 2: THE CRITICAL CONNECTION SEQUENCE

This must be done in the *exact* order.

SOLAR PANEL FOR YOUR RV



1. BATTERY → CHARGE CONTROLLER:

- **Action:** Connect the wires from the **Battery Bank** to the "**BATTERY**" terminals ($\oplus$ and $\ominus$) on the Charge Controller.
- **Why?** The controller *must* see the battery's voltage *first*. This allows it to boot up and "auto-detect" the system voltage (e.g., "Ah, this is a 24V system").
- Turn the **Battery-to-Controller breaker ON**. The controller screen should light up.

2. BATTERY → INVERTER:

- **Action:** Connect the main battery cables to the "**DC-IN**" terminals ($\oplus$ and $\ominus$) on the **Inverter**.
- **Why?** This powers up the inverter's "brain" and gets it ready.
- Turn the **Battery-to-Inverter breaker ON**. The inverter will beep and its screen will light up.

3. PV ARRAY → CHARGE CONTROLLER:

- **Action:** Connect the solar cables coming from the roof to the "**PV**" (or "**SOLAR**") terminals ($\oplus$ and $\ominus$) on the Charge Controller.
- **Why?** Now that the controller is *on* and *knows* it's a 24V system, it is safe to receive the high-voltage power from the panels.
- **DANGER:** If you connect the panels *first* (Step 3 before Step 1), the high voltage (e.g., 90V) from the panels will hit the "sleeping" controller, which doesn't know what voltage to regulate to. This can **instantly destroy** the controller.

4. INVERTER → AC LOADS:

- **Action:** With all DC components on, you can now turn **ON** the "**AC-OUT**" breaker from the inverter to the house.
- The house is now live.

SUMMARY: THE CONNECTION GOLDEN RULE

1. **Battery to Controller**
2. **Battery to Inverter**
3. **PV to Controller**
4. **AC to Loads**

Disconnection Sequence is the **EXACT REVERSE**:

1. **AC Loads (OFF)**
2. **PV (OFF)**
3. **Battery (OFF)**

EVALUATION

1. What is the *very first* electrical wire you should run, and where does it connect?
2. What is the **Critical Connection Sequence**? (List the 3 main DC steps in order).
3. Why is it *critical* to connect the **Battery** to the **Charge Controller** *before* connecting the **PV Array**?
4. What is the "Polarity" of a DC wire, and what are the standard colors for $\oplus$ and $\ominus$?

5. What is the correct **Disconnection Sequence**?

ASSIGNMENT

1. Write a 250-word "Safety Manual" for an apprentice, titled "The Critical Connection Sequence." Explain *why* the order (Battery, then PV) is so important and what happens if they get it wrong.
2. Draw a complete, labeled wiring diagram for the entire "Case Study" system from Week 6. It must show all 4 components (Panels, Battery, Controller, Inverter) and all 4 breakers (PV, Battery-to-Controller, Battery-to-Inverter, AC Out).
3. What does the term "Commissioning" mean? (This is a preview for next week).
4. Why are the battery cables usually the *thickest* cables in the entire system? (Hint: Review Week 8, 2nd Term - $P = V \times I$).
5. What is the main safety risk of a "loose connection" in a high-current DC circuit?



WEEK 10: SYSTEM TESTING & COMMISSIONING

INTRODUCTION

The system is fully installed, but the job is not finished. **Commissioning** is the formal, step-by-step process of **testing, verifying, and "handing over"** a new installation to the customer.

This is the final quality check to ensure the system is:

1. **SAFE:** No risk of shock or fire.
2. **CORRECT:** All components are wired as designed.
3. **PERFORMING:** The system is generating the power it is supposed to.

SOLAR SYSTEM COMMISSIONING CHECKLIST

System Verification & Testing

- ✓ Polarity Check 
- ✓ Wire Tightness 
- ✓ VOC (Open Circuit Voltage) Test
- ✓ Battery Voltage/SoC 
- ✓ Battery Voltage/SoC 
- ✓ AC Output Voltage/Frequency

Ensure safety & performance

FINAL STEPS



Handover to
Customer



Training &
Documentation
Provided



Professional, Thorough, and Safe Activation

STEP 1: PRE-POWER-ON CHECKS (VISUAL INSPECTION)

Before turning *any* breaker on, you must do a final visual check.

1. **POLARITY CHECK:** Check *every single* $\oplus$ and $\ominus$ connection. Is Red on $\oplus$? Is Black on $\ominus$? A reversed polarity will destroy the equipment.
2. **TIGHTNESS CHECK:** Use your screwdriver and spanner to check that *every* terminal and nut is tight. A **loose connection** creates high resistance, which creates **heat** and is a major **fire hazard**.
3. **SAFETY CHECK:** Are all ground wires connected? Are there any exposed copper wires? Is the battery area ventilated?
4. **DOCUMENTATION:** Have your Multimeter (DMM) and Clamp Meter ready.

STEP 2: COMMISSIONING (POWER-ON) & VOLTAGE/CURRENT TESTS

We follow the "Power-On" sequence from last week, but this time, we *test* at each step.

1. BATTERY BANK TEST:

- **Action:** Turn **ON** the **Battery-to-Controller** and **Battery-to-Inverter** breakers.
- **Test (with DMM):**
 - Measure the DC Voltage at the **Inverter's DC input**. (Should read $\approx 25.2\text{V}$ for our 24V bank).
 - Measure the DC Voltage at the **Controller's Battery input**. (Should read $\approx 25.2\text{V}$).
- **Result:** The Inverter and Controller screens should light up.

2. PV ARRAY TEST:

- **Action:** Go outside. *Before* turning on the PV breaker, measure the "open" string.
- **Test (with DMM):**
 - Measure the **Array V_{OC} (Open Circuit Voltage)** at the PV breaker (while it's OFF).
 - (e.g., For our 3 x 46V V_{OC} panels in *parallel* - oops, bad design in Wk 6, let's fix that. In our Wk 6 case study, we had 3x 330W panels. We'd connect them in *series*).

- **Corrected Wk 6 Design:** 3 x 330W panels. $V_{OC} \approx 46V$. **Connect in Series.**
- **Test:** $V_{OC} = 46V + 46V + 46V = 138V$.
- **Verify:** Is 138V *less than* our controller's 150V rating? **Yes.** It is safe.
- **Action:** Now, turn **ON** the **PV-to-Controller** breaker.

3. PERFORMANCE CHECK (CHARGING):

- **Test (with Clamp Meter):**
 - Clamp the meter around the $\oplus$ wire coming from the **PV Array**. (e.g., It might read **8.1A**).
 - Clamp the meter around the $\oplus$ wire going to the **Battery**. (e.g., It might read **29.5A**).
- **Verify:** This *proves* the **MPPT controller** is working! It is converting the high-voltage/low-current input (e.g., $990W / 122V = 8.1A$) into the low-voltage/high-current output for the battery (e.g., $990W / 27V = 36.6A$ - *this is the $I = P/V$ calculation*).
- **Result:** The controller's screen should show it is in "MPPT" or "Bulk" charging mode.

4. AC OUTPUT TEST:

- **Action:** Turn **ON** the **AC-Out** breaker.
- **Test (with DMM):**
 - Set DMM to V_{AC} .
 - Measure the voltage at the wall sockets.
- **Verify:** The reading should be within the correct range for Nigeria (e.g., **220V - 240V**).
- **Test:** Plug in a load (e.g., a fan) and ensure it runs smoothly.

STEP 3: HANDOVER TO CUSTOMER

The final step is to *train the customer*.

1. **Explain the System:** Show them the main components (panels, inverter, batteries) and what they do.
2. **Explain the "Normal" State:** Show them what the lights on the inverter/controller mean (e.g., "Green light means charging," "Red light means fault").
3. **Teach the "On/Off" Sequence:** Show them the 3-4 breakers and the correct sequence to shut down and restart the system.
4. **Provide a Maintenance Schedule:**
 - **Monthly:** Clean dust/dirt off the panels (early morning, with water and a soft cloth).
 - **Bi-Annually:** Check battery water levels (if flooded lead-acid).
 - **As-Needed:** Call the installer if any error codes appear.

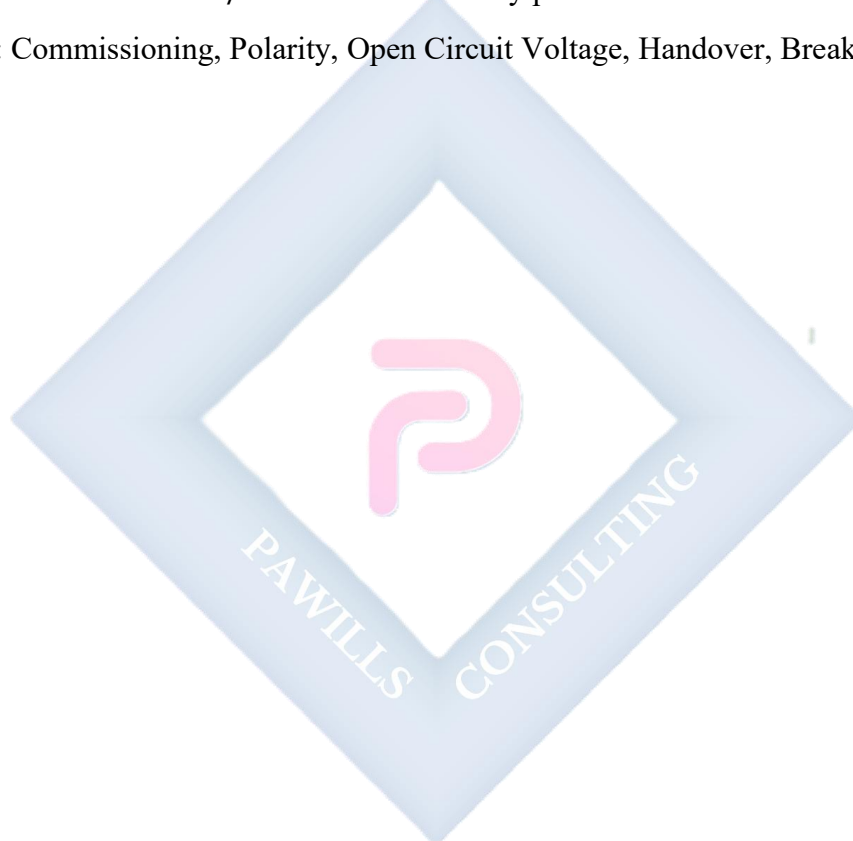
EVALUATION

1. What is "Commissioning"?
2. What is the *first* test you should perform *before* turning on any breakers? (Hint: Visual...).
3. What is the correct 3-step *power-on sequence* for commissioning?
4. What tool would you use to check the **AC Voltage** at a socket, and what reading would you expect?
5. What two key things must you teach the customer during the "Handover"?

ASSIGNMENT

1. Create a "System Commissioning Checklist." This should be a 1-page form a technician would fill out. It must include sections for:
 - (a) Visual Checks (Polarity, Tightness)
 - (b) V_{OC} Test (Expected vs. Actual)
 - (c) $V_{battery}$ Test (Actual)

- (d) V_{AC_out} Test (Actual)
 - (e) A final "System Live" signature box.
2. Write a 200-word "Handover Script" (what you would say to the customer) explaining how to clean their solar panels safely.
 3. An installer uses a DMM to test a 24V battery bank. The reading is 19.5V. What does this reading tell you? (Is the system healthy?).
 4. An installer uses a Clamp Meter on a panel string in full sun and reads 0.1A. His Solarimeter reads 900 W/m^2 . What is the likely problem?
 5. Define: Commissioning, Polarity, Open Circuit Voltage, Handover, Breaker.



WEEK 11-13: REVISION & EXAMS

GENERAL REVIEW OF THE THIRD TERM'S TOPICS

This final period is for a comprehensive review of the entire System Design and Installation process. We have moved from basic theory to a complete, practical installation.

KEY TOPICS REVIEW

- **WEEK 1-6: SYSTEM DESIGN & SIZING (THE "MATH")**
 - **Step 1: Load Assessment (W1):** The foundation. Calculating **Total Daily Energy (Wh/Day)** from the customer's appliance list (Watts × Hours).
 - **Step 2: Battery Sizing (W3):**
 - $(\text{Wh/Day} \div \text{Inverter Eff.}) \times \text{Days of Autonomy} = \text{Usable Wh}$
 - $\text{Usable Wh} \div \text{DoD (0.5 for Lead-Acid, 0.9 for Lithium)} = \text{Total Wh}$
 - $\text{Total Wh} \div \text{System Voltage (12/24/48V)} = \text{Total Ah}$
 - **Step 3: Array Sizing (W2):**
 - $(\text{Wh/Day from Step 2.1}) \div \text{PSH (Peak Sun Hours)} = \text{Raw Watts}$
 - $\text{Raw Watts} \times \text{Derating Factor } (\approx 1.3) = \text{Final Watts } (W_p)$
 - **Step 4: Inverter Sizing (W4):**
 - **Voltage:** Must match the battery (12/24/48V).
 - **Waveform:** Must be **Pure Sine Wave**.
 - **Power (W):** Must be $> (\text{Total Continuous Load} \times 1.25)$ AND $> (\text{Total Surge Load})$.
 - **Step 5: Controller Sizing (W4):**
 - **Type:** Must be **MPPT** if $\text{Panel V} > \text{Battery V}$.
 - **Voltage (V_{OC}):** Must be $> V_{OC}$ of the panel string.
 - **Current (A):** $(\text{Total Panel } W_p / \text{Battery V}) \times 1.25$.
- **WEEK 8-10: SYSTEM INSTALLATION & TESTING (THE "PRACTICAL")**
 - **Week 8: Mounting:** Installing mounts (pitched vs. flat), rails, and the **BOS** (Balance of System) on the component board.

- **Week 9: Wiring Sequence (CRITICAL):**
 - i. **Battery → Controller** (So it can "see" the voltage)
 - ii. **Battery → Inverter**
 - iii. **PV → Controller** (Last DC step)
 - iv. **AC Out → Loads**
- **Week 10: Commissioning:** The formal power-on sequence (same as above) using **Multimeters** and **Clamp Meters** to *verify* all voltages and currents are correct *before* handing over to the customer.

FINAL EXAM

The final examination will test all aspects of this term's work, especially the **Sizing Calculations (Weeks 1-6)** and the **Installation Safety/Sequencing (Weeks 8-10)**. Be prepared to design a complete system from a given load list.

